

Changelog for GPW WATS 5.01 Risk Management Gateway

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1. DISCLAIMER

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In case of sections of documentation at a **High** level work progress according to the current version of *GPW WATS Advancement of Documentation*, Warsaw Stock Exchange will endeavor to limit changes to these sections of documents to those related to:

1. correcting errors in the documentation or in the software;
2. clarification of the documentation content or removing ambiguity;
3. implementation of approved change requests or;
4. regulatory changes.

2. PREFACE

This document has been prepared by Warsaw Stock Exchange in order to help in the implementation process of GPW WATS trading platform. This section describes the basic information about Risk Management Gateway. You can learn about the target audience, the document purpose and all the necessary documents you should read in relation to this Specification.

2.1. TARGET AUDIENCE

This document has been prepared to development staff, Independent Software Vendors who produce software integrated with GPW WATS, analysts, market participants and all clients who want to deepen their knowledge about GPW WATS.

2.2. DOCUMENT PURPOSE

This document presents all adjustments that has been made in the newest release. The document is always compared with its previous version. Comparison of format changes is not included for readability purposes.

2.3. ASSOCIATED DOCUMENTS

GPW WATS 5.01 Risk Management Gateway is a part of GPW WATS documentation set.

Please check the following documents to learn about the construction of Trading System.

- GPW WATS 1.01 Trading System.

Please check the documentation of the trading protocols supported by GPW WATS.

- GPW WATS 2.01 Native Order Gateway Specification,
- GPW WATS 2.02 FIX Order Gateway Specification.

Please check the description of the communication with Data Distribution Service.

- GPW WATS 3.01 Market Data Protocol.

Please check the additional documentation which explains other services provided within GPW WATS.

- GPW WATS 4.01 Drop Copy Gateway,
- GPW WATS 4.02 Post Trade Gateway,
- GPW WATS 5.01 Risk Management Gateway (this document).

Please check the additional documentation describing the following:

- GPW WATS 2.03 Rejection Codes,
- GPW WATS 2.04 BenDec Message Definition Format,
- GPW WATS 6.01 Connectivity.

It is recommended that you read the GPW WATS 1.01 Trading System document first.

3. DOCUMENT HISTORY

version	Date	Description
0.51	29.06.2023	The initial publication of the documentation.
0.52	26.07.2023	Addition of a new risk limit type to FIX protocol. Messages updated: CS, CT, CL, CM - Tag: RiskLimitType - Value: RISK_LIMIT_NOT_DEFINED
0.53	28.08.2023	Improved the readability of the Order Value Control tables in Section 5.3 and Chapter 6. Updated RM protocol message descriptions in Chapter 8 RM Protocol.
0.53*	30.09.2023	RM Protocol - Update of FIX RM protocol messages - Addition of Native RM protocol New Sections: - Section 7.2 Breach Action Message Flow in Pre-Trade Controls - Sections 7.2.1 and 7.2.2 Kinematics diagrams illustrating pre-trade control action - Section 7.3 RM Parametrization - Section 7.4 RM Controls Configuration - Section 8.3 Concordance - Section 8.4 RM Control of Trading Activity - Section 8.5 Order Identification - Section 8.6 Pre-Trade checks and Order Types - Section 8.7 Order Calculation
0.56	17.11.2023	RM Protocol - Update of FIX RM protocol messages - Removal of Native RM protocol New Sections: - Section 8.1 FIX RM Protocol Messages - Section 8.2 FIX RM Protocol Message Examples
0.57	27.11.2023	RM Protocol - Section 8.1.2 New error codes for the PartyRiskLimitsDefinitionRequestAck[CT] message. The RiskLimitRequestResult [1761] attribute is updated. Updated Sections: - Section 7.4.5 Error Codes. Update to the descriptions of error codes.
0.58	15.12.2023	Publication of v0.58.
0.59	20.01.2024	Publication of v0.59.
0.60	20.02.2024	Minor changes to the document. Section 4.3 incorporates updated example market structure. Native protocol Risk Management configuration functionality is deprecated. Native RM messages removed from the system are: - RiskLimitBreach - RiskLimitDefinition - RiskLimitDefinitionResponse From now on, Risk Management configuration is exclusively carried out by means of FIX protocol messages CS, CT, CL and CM. References to Native Risk Management messages have been removed from the document. MassQuote monitoring is now part of RM. Sections 4.3.3, 4.4 and 7.4.1 updated accordingly.
0.61	29.02.2024	Added Section 8.3 Message Kinematics which describes the kinematics of RM messages: - PartyRiskLimitsDefinitionRequest (CS) - PartyRiskLimitsDefinitionRequestAck (CT) - PartyRiskLimitsRequest (CL) - PartyRiskLimitsReport (CM)

version	Date	Description
0.62	25.03.2024	Publication of v0.62. No changes in this version.

4. GPW WATS RISK MANAGEMENT GATEWAY

4.1. INTRODUCTION

GPW WATS Risk Management Access (RM) is a set of Exchange Level Controls which offer a risk-management capability to Exchange Members and Sponsored Access users. This functionality, available to Risk Entities, covers all financial instruments. Exchange members as well as clearing members in their role of Risk Controllers, where they can define risk controls for Market Participants including Sponsored Access users, monitor risk levels and receive breach of trading limit notifications.

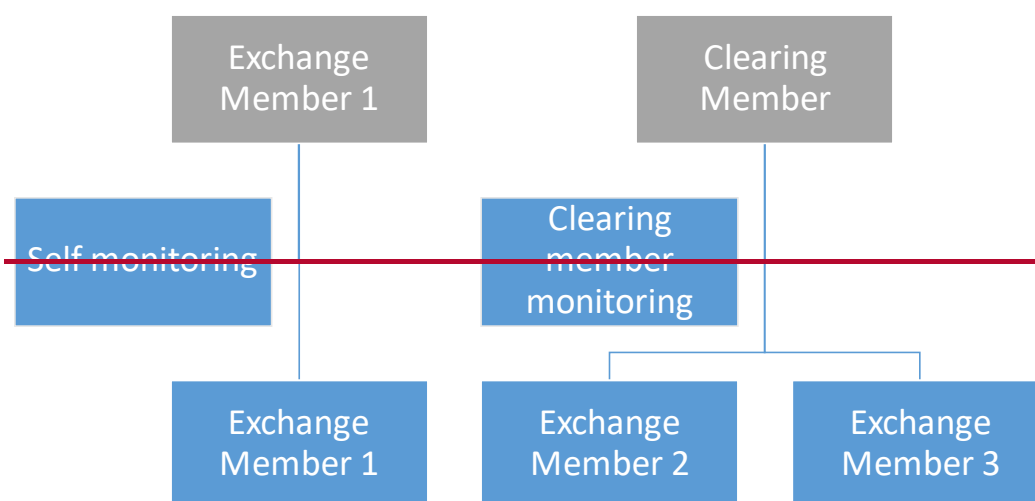
GPW WATS RM implements pre-trade and post-trade risk controls. The pre-trade controls fulfill the requirements of MIFID II/MIFIR under RTS 7 Article 20 Pre-trade and post-trade controls (Article 48(4) and (6) of the Directive 2014/65/EU).

4.2. USERS AND ROLES

Risk management within GPW WATS RM is carried out by the risk manager. Risk managers are either trading participants (Exchange Members) or clearing firms who offer clearing services to market participants. Risk managers configure limits, receive limit breach notifications and monitor risk usage. Clearing members may manage risks for multiple exchange members and exchange members may have one or more clearing firms as risk managers.

RM User type	Characteristic	Risk Management Role
Exchange Members (EM)	EMs are market (trading) participants in one of the following capacities: Agency Principal Market Maker Institutional clients	EMs can configure risk controls for their clients including Sponsored Access clients and Principal / Market Maker activity
Clearing Members (CM)	CMs offer clearing and risk management services to Exchange Members	CMs can configure risk controls for those EMs with whom they have a clearing or risk management relationship

Risk Management Function	Description.
Set and modify risk limits	Risk managers configure RM settings for their clients or themselves
View current risk limits and risk limit usage	Risk managers view current risk limits and usage
Receive notifications of risk limit usage	Risk managers may receive notifications

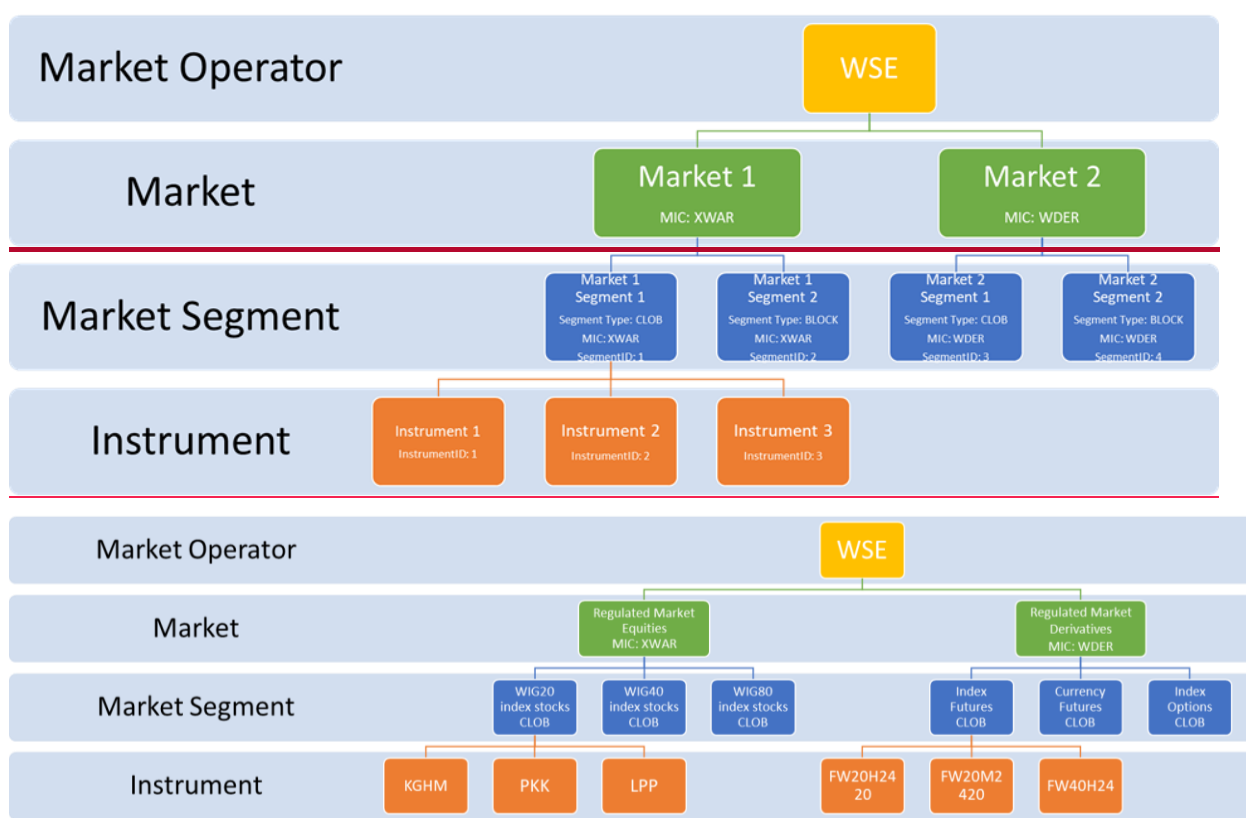
Figure 1 – Risk Monitoring Schemes

RM-User type	Risk-Management Rights
Exchange Members	Exchange and Clearing members have the same risk management rights
Clearing Members	Configure risk limits Monitor risk limit usage

4.3. RM CONFIGURATION STRUCTURE AND HIERARCHY

In GPW WATS RM, risk managers may configure risk controls at various levels of the market structure: at the level of market, market segment or individual financial instrument. An example market structure to help illustrate the possible risk configurations is presented below:

Figure 1 - Example Market Structure



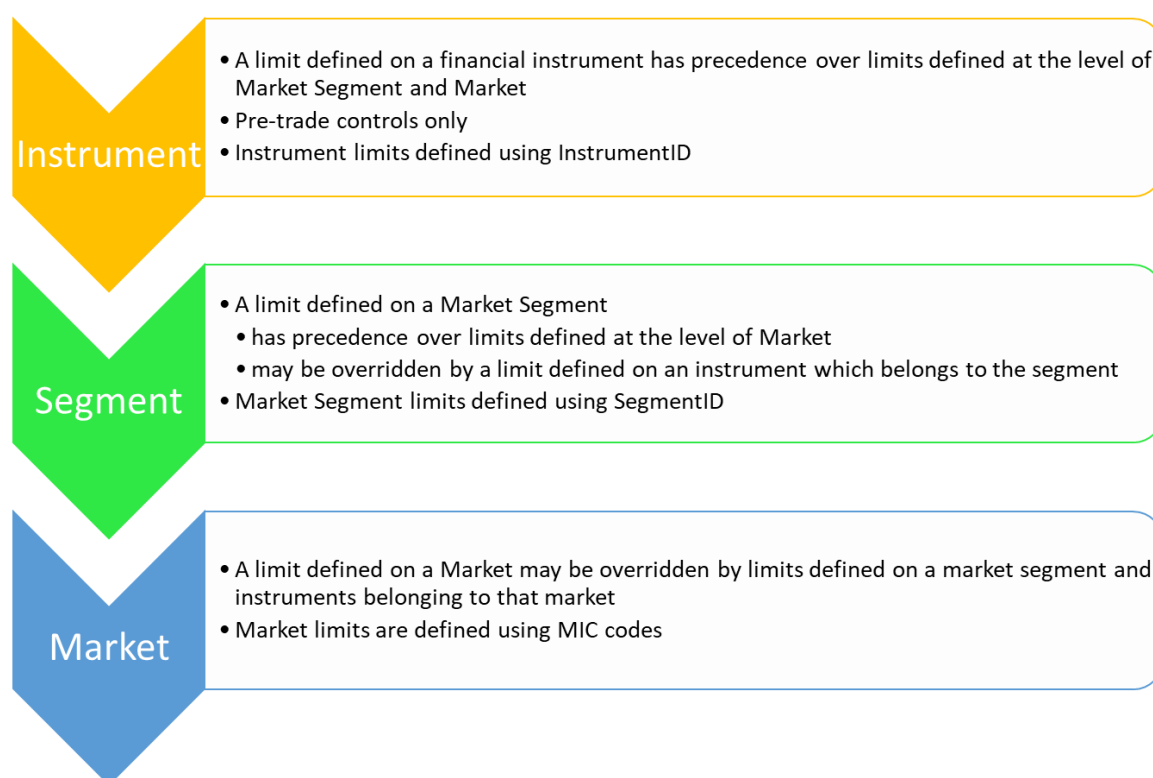
The following table presents details of the mechanisms for configuring RM limits across the market structure.

Hierarchy	Level	ID	Description
1	Instrument	InstrumentID	Limits may be configured at the level of an individual financial instrument using InstrumentID.
2	Market Segment	SegmentID	An entire market segment may be configured using a SegmentID. The same limit will apply across all instruments listed in that segment.
3	Market	MIC code	A limit may be applied to an entire market. The limit will apply to all market segments and instruments. Venues are identified by MIC code.

A hierarchy of risk limits is adhered to in RM. Limits defined are applied in the following order of precedence: financial instruments, market segments and markets. This design enables risk managers to define "fine-grained" risk controls to meet the needs of their clients.

NOTE: POST-TRADE CHECKS CANNOT BE APPLIED AT THE FINANCIAL INSTRUMENT LEVEL.

Figure 2 - Market Structure and Risk Limit Definition



Example 1: a limit defined on a financial instrument (InstrumentID 1) will supersede limits defined on the market segment which lists that instrument and the limit set at the level of market.

Example 2: a limit set at the level of market segment (SegmentID 2) will apply to all securities in that segment. It will also have precedence over a limit set at the level of a market hosting that segment.

4.3.1. DEFAULT CONTROLS

Risk management default behavior differs between pre-trade and post-trade controls. Pre-trade controls are mandatory and require the definition of control limits. If any of the 3 pre-trade controls are undefined, all orders will be automatically rejected. For post-trade checks, the behavior is reversed since post-trade controls are not mandatory. When a post-trade control is undefined, the order is accepted and routed into the trading engine.

Control type	Default behavior	Note
Pre-trade	Reject	If any pre-trade control is undefined, all orders entered are rejected by the system.
Post-trade	Pass-through	If post-trade controls are undefined, all orders entered in the system are accepted.

4.3.2. CLEARING MEMBER CONFIGURATION AND RISK MANAGEMENT

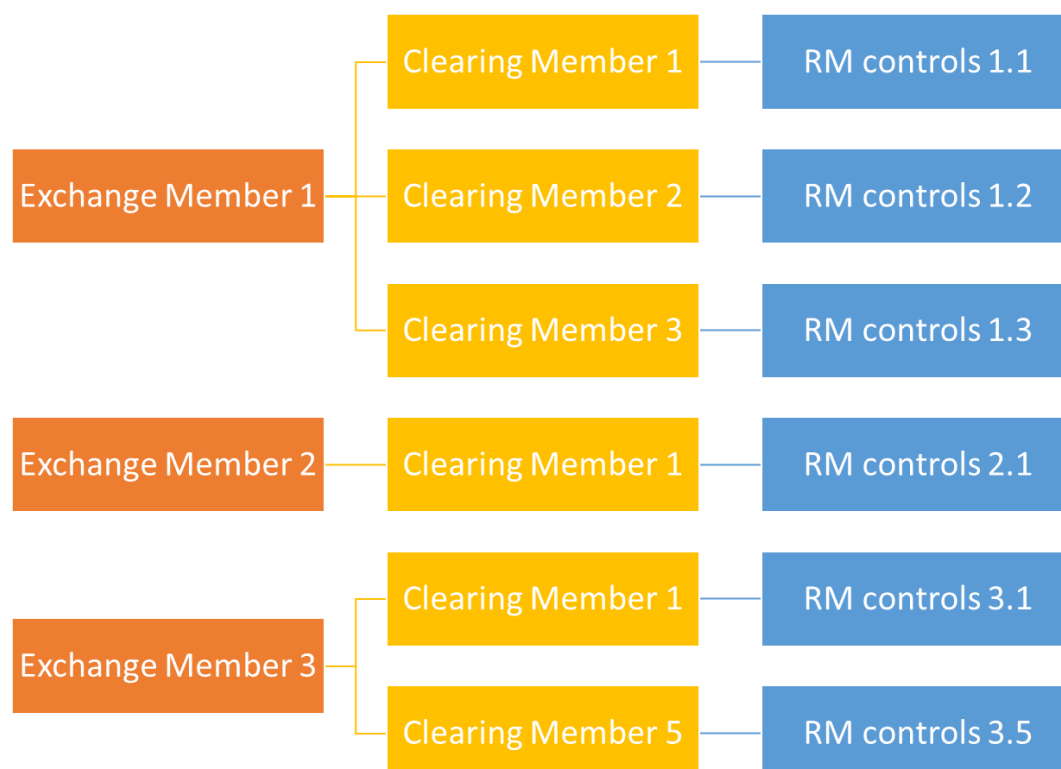
GPW WATS RM enables clearing members to manage the risk related to orders sent by the trading member for which a given clearing member offers a clearing service.

Within RM, a clearing member is a risk manager for a trading member. At the same time, RM enables several risk managers (clearing members) to control the orders of a single trading member.

The mechanism to identify how a specific order should be processed by the RM is an integral part of all orders submitted to the system. All orders submitted by the Native or FIX Order Gateway include information which indicates the relevant clearing member and the relevant risk controls to be utilized.

The diagram below illustrates the one-to-many relationship between a trading member and its risk managers. In the example, Exchange member 1 has 3 clearing members who act as risk managers each configure a separate set of risk controls for trades they are responsible for clearing. Exchange members 2 and 3 share a clearing member with exchange member 1 who controls their risk settings. Each set of risk controls is configured individually and separately.

Figure 3 - Relationship between Members and Risk Limits



4.3.3. RM MONITORED OPERATIONS

RM monitors order entry and order modification operations by exchange members on both the FIX and Native Order Gateways. Order messages are screened by the RM system and either passed onto the trading engine or rejected. The following table lists the order messages screened by GPW WATS RM together with the message fields relevant to computing the risk metrics and applying risk controls.

Message Type	Incoming order message	Message tags relevant to risk control
Order entry	FIX: NewOrderSingle [type 'D']	Price [44] – the price specified on the order OrderQty [38] –order quantity OrderCapacity (528) – capacity of the Trading Firm OrderAttributeGrp (2593) - liquidity provision activity order mark.
	Native: OrderAdd	price – the price specified on the order quantity - order quantity capacity – capacity of the Trading Firm

Message Type	Incoming order message	Message tags relevant to risk control
		mifidFields = liquidity provision activity order mark.
Order modification	-FIX: OrderCancelReplaceRequest [type 'G']	Price [44] – the price specified on the order OrderQty [38] – order quantity
	Native: OrderModify	price – the price specified on the order quantity - order quantity

4.3.4. IDENTIFICATION OF CLEARING MEMBER

Each order entered into GPW WATS features information about the Clearing Member responsible for clearing the related transaction. This information is also used to identify and apply risk controls. The mechanism for to identify orders is outlined in the table below:

Protocol	Quote entry	Order Entry Instruction	FIX: MassQuote [type 'i']	Clearing Member Identification BidPx [132], OfferPx [133] – quote price BidSize [134], OfferSize [135] – quote size	Note
FIX		Native: MassQuote	bid.price, ask.price – quote price bid.quantity, ask.quantity – quote size		
Quote modification		A quote may be modified by overwriting an existing quote in the system using a FIX or Native port quote entry message.			

4.3.5. 4.3.4. CONFIGURABLE CONTROLS

The risk controls configuration consists of setting the appropriate limits on operations performed by exchange members. These limits consist of maximum values on certain variables related to order entry and modification. The limits are expressed in terms of currency, number of securities (instrument volume) and other types and are presented in the table below.

Control	Control name	Limit parameter	Note
Pre-trade	Order price collars	Multiplier	The multiplier specifies the extent to which order price collars can stretch relative to static price collars.is Type: integer, between 10% and 99%.
	Maximum order value	MaxOrderValueBuy MaxOrderValueSell	Maximum value of an order. Distinct limits for buy and sell sides. Type: currency value, integer.
	Maximum order volume	MaxOrderVolumeBuy MaxOrderVolumeSell	The maximum order volume, in units of a financial instrument. Distinct values for buy and sell side. Type: number of units, integer (lot size * order quantity).
	Restricted instrument list	List of restricted stocks	A list which consists of InstrumentIDs in which the EM is not allowed to trade
Post-trade	Traded Value Controls	TotalTradedValueLimit TotalTradedBuyValueLimit TotalTradedSellValueLimit	Limits on the total value of transactions performed by an EM. Separate limits for buy, sell side and total trades. Type: currency value, integer.
	Open Orders Value Controls	TotalOpenBuyValueLimit TotalOpenSellValueLimit TotalOpenValueLimit	Limits on the total value of open orders. Separate limits for buy, sell side and total orders. Type: currency value, integer.
	Risk Value Controls	TotalBuyRiskValueLimit TotalSellRiskValueLimit TotalRiskValueLimit TotalNetRiskValueLimit	Limits on the total value and net value of open orders and transactions. Separate limits for buy, sell side. Type: currency value, integer.

Control	Control name	Limit parameter	Note
	Order Count Controls	MaxOrderCount	The maximum number of order messages an EM is allowed to submit during the trading day.

4.4. SCOPE OF RM CONTROLS

All orders entered into GPW WATS by means of FIX and Native Order Gateway messages as outlined in the following table are screened and controlled by the RM subsystem:

Message Type	FIX Order Gateway	Native Order Gateway
Order entry	NewOrderSingle [type 'D'], <u>MassQuote [type 'I']</u>	OrderAdd, <u>MassQuote</u>
Order modification	OrderCancelReplaceRequest [type 'G'], <u>MassQuote [type 'I']</u>	OrderModify, <u>MassQuote</u>

4.4.1. RM AND TRADING PHASES

Interaction with the RM system, including the setting of limits, interrogation of limit bounds is allowed at all times.

RM makes no distinction between Trading Phases and applies the same controls across all phases during the trading day. There is no facility to configure different RM controls by trading phase such as continuous trading or auction.

4.4.2. GPW WATS ORDER FLOWS

GPW WATS FIX and Native Order Gateway order messages can be formatted to convey relevant capacity information in both the FIX and Native protocols. This section presents the FIX and Native protocol order entry and modification messages together with the message fields and values required to encode the capacity of an order.

GPW WATS RM recognizes 4 distinct capacities in which an exchange member may submit orders:

5. Agency – all orders submitted by a member's clients
6. Principal/Dealer – an exchange member's principal/dealer activity on their own account
7. Market making activity or an exchange member
8. Institutional client order flow – the order flow of selected institutional clients

In this section we explain how GPW WATS identifies various order flows and how these order flows should be identified and tagged in orders submitted using the FIX and Native protocols. The different capacities supported by GPW WATS RM will be enumerated and the ways in which orders should be identified will be explained.

GPW WATS supports 2 trading interfaces: a FIX trading interface and a Native trading interface. Each of these interfaces support distinct messages for order entry and order modification. In the FIX protocol, the GPW WATS FIX Order Gateway supports order entry and modification using the NewOrderSingle [type 'D'] and OrderCancelReplaceRequest [type 'G'] messages. The Native Order Gateway on the other hand implements the OrderAdd and OrderModify messages. Each protocol supports a different syntax, and this section will provide clear examples on how orders with different capacities may be submitted in using each protocol.

4.4.3. FORMAT OF TRADING PORT ORDER MESSAGES

4.4.3.1. Agency

Agency orders relate to all orders submitted by a member's clients. In FIX they are identified by the OrderCapacity [528] tag carrying the value A. In the Native protocol, the capacity field of the OrderAdd message set to 1 (Agency) identifies an order as an agency order. The remaining fields are left empty.

Case 1: Identification of all client orders of a specific exchange member excluding clients engaged in market making.

	FIX Order Gateway		Native Order Gateway	
FIX Field Name	FIX Tag	NewOrderSingle [type 'D']	Field name	OrderAdd
OrderCapacity	528	A = Agency	capacity	1 = Agency
OrderAttributeType	2594	<u>2 = Liquidity provision activity order</u>		
OrderAttributeValue	2595	<u>N</u>		
PartyID	448			
PartyIDSource	447			
PartyRole	452			
PartyRoleQualifier	2376			

Case 2: Identification of orders of clients solely engaged in market making.

	FIX Order Gateway		Native Order Gateway	
FIX Field Name	FIX Tag	NewOrderSingle [type 'D']	Field name	OrderAdd
OrderCapacity	528	A = Agency	capacity	1 = Agency
OrderAttributeType	<u>2594</u>	<u>2 = Liquidity provision activity order</u>		
OrderAttributeValue	<u>2595</u>	<u>Y</u>		
PartyID	<u>448</u>			
PartyIDSource	<u>447</u>			
PartyRole	<u>452</u>			
PartyRoleQualifier	<u>2376</u>			

4.4.3.2. Principal/Dealer

The principal/dealer capacity identifies orders entered by a firm on its own account. In FIX these are identified by OrderCapacity [528] set to P. Native protocol order messages require the capacity field to be set to 2 (Principal).

Order entry		FIX Order Gateway	Native Order Gateway	
FIX Field Name	FIX Tag	NewOrderSingle [type 'D']	OrderAdd	Field value
OrderCapacity	528	P = Principal	capacity	2 = Principal
OrderAttributeType	2594	n/a 2 = Liquidity provision activity order		
OrderAttributeValue	2595	n/a N		
PartyID	448			
PartyIDSource	447			
PartyRole	452			
PartyRoleQualifier	2376			

4.4.3.3. Market Maker

Orders entered in a Market Maker capacity are identified in FIX by OrderCapacity [528] set to P and associated tags as in the table below. Native protocol order messages require the capacity field to be set to 2.

Order entry		FIX Order Gateway	Native Order Gateway	
FIX Field Name	FIX Tag	NewOrderSingle [type 'D']	OrderAdd	Field value
OrderCapacity	528	P = Principal	capacity	2 = Principal
OrderAttributeType	2594	2 = Liquidity provision activity order	mifidFields.flags	0b0001 - LiquidityProvisionActivity
OrderAttributeValue	2595	Y		
PartyID	448			
PartyIDSource	447			
PartyRole	452			
PartyRoleQualifier	2376			

4.4.3.4. Institutional Clients

RM can apply personalized filters to individual institutional clients. For this purpose, the institutional client needs to be identified in all orders submitted in the FIX and Native protocols. Tagging an institutional client requires the OrderCapacity [528] to be set to A (agency) and supplying the Short Code of the institutional client, whose orders are to be controlled by GPW WATS RM. In addition, PartyRoleQualifier [2376] needs to be set to 23 to identify that the order comes from a legal entity.

In the Native Order Gateway, the same information needs to be provided in the relevant fields.

Order entry		FIX Order Gateway	Native Order Gateway	
FIX Field Name	FIX Tag	NewOrderSingle [type 'D']	OrderAdd	Field value
OrderCapacity	528	A = Agency	capacity	1 = Agency
OrderAttributeType	2594	n/a 2 = Liquidity provision activity order		
OrderAttributeValue	2595	n/a N		
PartyID	448	Short Code identifying the institutional client	mifidFields.client.shortCode	shortCode value
PartyIDSource	447	P = Short code identifier		
PartyRole	452	3 = Client ID		
PartyRoleQualifier	2376	23 = Firm or legal entity (LEI)	mifidFields.client.qualifier	3 - FirmOrLegalEntity

In the case the institutional client is engaged in market making activities, the tags 2594 and 2595 must be filled as follows:

FIX Field Name	FIX Tag	Value
OrderAttributeType	2594	2 = Liquidity provision activity order
OrderAttributeValue	2595	Y

5. PRE-TRADE RISK CONTROLS

5.1. RESTRICTED INSTRUMENT LIST

A Restricted Instrument List is a list of financial instruments in which an Exchange Member is not allowed to trade. Any order messages entered for instruments on that list are rejected.

Control Type: Pre-trade check

Market Applicability: ~~All Financial Instruments~~

Equities, Bonds, Derivatives **Breach Action:** reject order on arrival to the FIX or Native Order Gateway if an order is placed for an instrument on the restricted list. The following table shows rejection messages in response to affected incoming order messages:

Incoming order message	Rejection message
NewOrderSingle [type 'D']	Execution Report (8), with fields ExecType [150]: "8 = Rejected"
OrderCancelReplaceRequest [type 'G']	OrderCancelReject [type 'g'], with fields OrdStatus [39]: "8 = Rejected"
OrderAdd	OrderAddResponse status = Rejected
OrderModify	OrderModifyResponse status = Rejected

Order types: Market, Limit, Iceberg, Stop and modifications.

Scope of limit: The limit can be set at the level of Market (via MIC code), market segment or individual financial instrument.

Default limit: list is empty, EM can enter orders for any instrument traded on Market.

System level limit: none

Calculation: check if the order placed for instrument is on the list

Check application:

FIX Order Gateway		
Order type	Behavior	Control Formula
Buy order	check if SecurityID[44] on list, if yes reject order	SecurityID[44] on list
Sell order		SecurityID[44] on list

Native Order Gateway		
Order type	Behavior	Formula
Buy order	check if <i>instrumentId</i> field in OrderAdd message on list, if yes reject order	<i>instrumentId</i> on list
Sell order		<i>instrumentId</i> on list

Configuration: Restricted instrument lists are defined by the responsible Clearing member. Restricted Instruments List is configured at RM parametrization level. They consist of InstrumentIDs of selected financial instruments, SegmentIDs of market segments or MICs

Design: Part of the standard order routing. Each incoming order is checked against the list and rejected if it fails the check.

Relevant regulation: not applicable

5.2. ORDER PRICE CONTROL

RM Order Price Control ensures orders fit within reasonable price bands before entering the order book. GPW WATS implements separate order price collars for buy and sell orders. The limits are configured as a multiple of the static price collar range defined for a given financial instrument ~~relative to the reference price~~.

Control Type: Pre-trade check

Market Applicability: ~~All Financial Instruments~~ Equities, Bonds, Derivatives

Breach Action: reject order on arrival to the FIX or Native Order Gateway if the order price out-of-bounds of respective buy or sell RM price collars. The following table shows rejection messages in response to affected incoming order messages:

Incoming order message	Rejection message
NewOrderSingle [type 'D']	Execution Report (8), with fields ExecType [150]: "8 = Rejected" OrdRejReason [103]: "3=Order exceeds limit"
OrderCancelReplaceRequest [type 'G']	OrderCancelReject [type 'g'], with fields OrdStatus [39]: "8 = Rejected" CxlRejReason [102]: "8=Price exceeds current price band"
OrderAdd	OrderAddResponse status = Rejected
OrderModify	OrderModifyResponse status = Rejected

Order types: Market, Limit, Iceberg, Stop and modifications.

Scope of limit: The limit can be set at the level of Market (via MIC code), market segment or individual financial instrument.

Default limit: GPW WATS static order price collars

System level limit: GPW WATS static order price collars

Calculation: The calculation of the RM order price collars is illustrated in the following table:

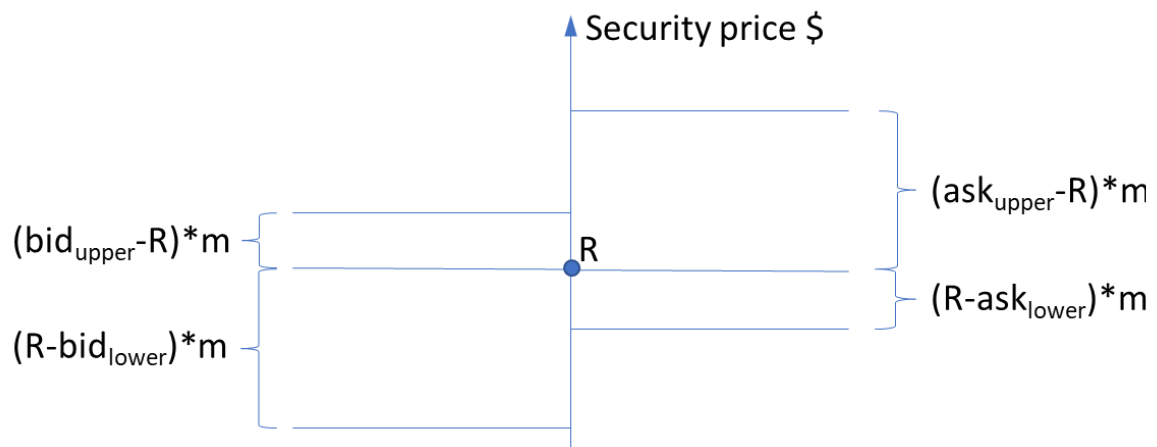
Order Type	RM Collar Bound	Formula
Buy	RMUpperBid	$R + (\text{bidupper} - R) \cdot m$
Buy	RMLowerBid	$R - (R - \text{bidlower}) \cdot m$
Sell	RMUpperAsk	$R + (\text{askupper} - R) \cdot m$
Sell	RMLowerAsk	$R - (R - \text{asklower}) \cdot m$

Notation:

R: static collar reference price;

Order Type	RM Collar Bound	Formula
Bid upper, bid lower, ask upper, ask lower : GPW WATS static order price collar values (upper and lower / for bid and ask separately), are communicated by the OrderCollars/TradeCollars/PriceUpdate Market Data message for any financial instrument m: multiplier in %; $-10\% < M < 99\%$		

RM Order Price collars for buy and sell orders



Check application:

FIX Order Gateway		
Order type	Behavior	Control Formula
Buy order	check if Price [44] in NewOrderSingle [type 'D'] message falls within configured RM price collars, reject order otherwards	$RMLowerBid \leq Price [44] \leq RMUpperBid$
Sell order		$RMLowerAsk \leq Price [44] \leq RMUpperAsk$
Buy order modification	check if Price [44] in OrderCancelReplaceRequest [type 'G'] message falls within configured RM price collars, reject order otherwards	$RMLowerBid \leq Price [44] \leq RMUpperBid$
Sell order modification		$RMLowerAsk \leq Price [44] \leq RMUpperAsk$

Native Order Gateway		
Order type	Behavior	Formula
Buy order	check if <i>price</i> field in OrderAdd message falls within configured RM price collars, reject order otherwards	$RMLowerBid \leq price \leq RMUpperBid$
Sell order		$RMLowerAsk \leq price \leq RMUpperAsk$
Buy order modification	check if <i>price</i> field in OrderModify message falls within configured RM price collars, reject order otherwards	$RMLowerBid \leq price \leq RMUpperBid$
Sell order modification		$RMLowerAsk \leq price \leq RMUpperAsk$

Configuration: RM order price collars are defined in relation to GPW WATS static order price collars configured in the system for a given tradable security. Collars are set through a Multiplier parameter which determines the spread of the price bands around the reference price.

Design: Part of the standard order routing. Each incoming order is checked against the collars and rejected if it fails the check.

Relevant regulation: This pre-trade check is implemented to fulfill obligations under MIFID II, RTS 7 Article 20.

5.3. ORDER VALUE CONTROL

Value Control ensures that incoming orders with unusually large values are screened and are entered into the order book.

Control Type: Pre-trade check

Market Applicability: Equities, [Bonds](#), [Derivatives](#)

Breach Action: reject order on arrival to the FIX or Native Order Gateway if the order value is out-of-bounds of respective the RM limit. The following table shows rejection messages in response to affected incoming order messages:

Incoming order message	Rejection message
NewOrderSingle [type 'D']	Execution Report (8), with fields ExecType [150]: "8 - Rejected"
OrderCancelReplaceRequest [type 'G']	OrderCancelReject [type 'g'], with fields OrdStatus [39]: "8 - Rejected"
OrderAdd	OrderAddResponse status = Rejected
OrderModify	OrderModifyResponse status = Rejected

Order types: Market, Limit, Iceberg, Stop and modifications.

Scope of limit: The limit can be set at the level of Market (via MIC code), market segment or individual financial instrument.

Default limit:

System level limit:

Calculation: The calculation of RM value limits varies by type of instrument. Separate volume limits may be applied to buy and sell orders. Lot size is considered when computing order values.

Instrument type	Formula for calculating order value
Equities, ETFs and Structured Products	$\text{Price}^1 * \text{Volume}^2$
Bonds	$\text{Nominal} * \text{Volume}^2$
Futures	$\text{Price}^1 * \text{Multiplier} * \text{Volume}^2$
Options	$\text{Multiplier} * \text{Volume}^2 * \text{Strike Price}$

Notation:

- For non-priced buy and sell orders (Market Orders and Market to Limit Orders), the Upper Static Trade Price Collar is used as the Order Price.
 - If Static Trade Price Collar is not activated then Upper Static Order Price Collar is used.
 - If Static Order Price Collar is not activated then Upper Dynamic Trade Price Collar is used.
 - If Dynamic Trade Price Collar is not activated then Upper Dynamic Order Price Collar is used.
 - If no Collars are activated then Last Traded Price is used.
- Volume = Quantity (specified in the order) x Lot Size
- Multiplier for derivative calculations

Check application:

Behavior	Control Formula
Check if Order Value (calculated according to the formulas in the table above) is less than configured RM order value bound, reject order otherwards	Order Value <= RMValueBound

Configuration: Limit represented as a value in the currency of the traded instrument

Design: Part of standard order routing. Each incoming order is checked against the limit and rejected if it fails the pre-configured limit check.

Relevant regulation: This pre-trade check is implemented to fulfill obligations under MIFID II, RTS 7 Article 20.

5.4. ORDER VOLUME CONTROL

Volume Control ensures that incoming orders with unusually large values are screened and do enter the order book.

Control Type: Pre-trade check

Market Applicability: Equities, [Bonds](#), [Derivatives](#)

Breach Action: reject order on arrival to the FIX or Native Order Gateway if order volume out-of-bounds of respective RM limit. The following table shows rejection messages in response to affected incoming order messages:

Incoming order message	Rejection message
NewOrderSingle [type 'D']	Execution Report (8), with fields ExecType [150]: "8 = Rejected"
OrderCancelReplaceRequest [type 'G']	OrderCancelReject [type 'g'], with fields OrdStatus [39]: "8 = Rejected"
OrderAdd	OrderAddResponse status = Rejected
OrderModify	OrderModifyResponse status = Rejected

Order types: Market, Limit, Iceberg, Stop and modifications.

Scope of limit: The limit can be set at the level of Market (MIC code), market segment or individual financial instrument.

Default limit:

System level limit:

Calculation: Separate volume limits may be applied to buy and sell orders. Lot size is taken into account when computing order volumes.

Check application:

FIX Order Gateway		
Order type	Behavior	Control Formula
Buy order	check if OrderQty [38] in NewOrderSingle [type 'D'] message x LotSize is less than configured RM order volume bound, reject order otherwards	OrderQty [38] x LotSize <= RMVolumeBound
Sell order		OrderQty [38] x LotSize <= RMVolumeBound
Buy order modification	check if OrderQty [38] in OrderCancelReplaceRequest [type 'G'] message x LotSize is less than configured RM order volume bound, reject order otherwards	OrderQty [38] x LotSize <= RMVolumeBound
Sell order modification		OrderQty [38] x LotSize <= RMVolumeBound

Native Order Gateway		
Order type	Behavior	Formula
Buy order	check if <i>quantity</i> in OrderAdd message x LotSize is less than configured RM order value bound, reject order otherwards	Quantity x LotSize <= RMValueBound
Sell order		quantity x LotSize <= RMValueBound
Buy order modification	check if <i>quantity</i> in OrderModify message x LotSize is less than configured RM order value bound, reject order otherwards	quantity x LotSize <= RMValueBound
Sell order modification		quantity x LotSize <= RMValueBound

Configuration: Limit represented as volume in units of the financial instrument

Design: Part of the standard order routing. Each incoming order is checked against the limit and rejected if it fails the pre-configured limit check.

Relevant regulation: This pre-trade check is implemented to fulfill obligations under MIFID II, RTS 7 Article 20.

6. POST-TRADE RISK CONTROLS

6.1. INTRODUCTION

Post-trade risk controls manage aspects of a participant's trading activity over an entire trading session. Unlike pre-trade controls, post-trade controls are cumulative and monitor a participant's activity during a trading day.

RM distinguishes buy and sell side risk and computes aggregate statistics which capture a participant's total risk activity. Post-trade controls can therefore be used to shape the risk behavior of a trading member.

At the beginning of each trading day, the control limits are reset. During the trading session the controls are incremented in line with a participant's trading activity. When a participant exceeds a given control limit, a range of actions are available from notification to blocking trading.

Unlike RM pre-trade checks, RM post-trade controls do not pre-validate incoming orders. Under the post-trade scheme, order entry, modification and cancellation are allowed unless risk limits are exceeded. Breach actions are only undertaken when post-trade limits are exceeded. Example: A user has a configured maximum Total Risk Value limit of \$1,000,000. He first submits a limit buy order for a value of \$200,000. He then submits a market buy order for a value of \$1,000,000. RM post-trade controls will not block the second order because they do not perform order pre-validation, and at the time of entry, the post-trade limit was not yet exhausted. The second order exhausts the post-trade limit configured for the user. Entry of a third order will not be allowed and any subsequent order will be rejected, unless the configured limit is adjusted.

GPW WATS RM implements the following post-trade controls:

Category	Risk Measure	Trading Side Risk Measure	Definition
Traded Value Controls	Total Traded Value		Total Traded Buy Value + Total Traded Sell Value
		Total Traded Buy Value	The value of buy transactions during a trading day.
		Total Traded Sell Value	The value of sell transactions during a trading day.
Open Orders Value Controls	Total Open Orders Value		Total Buy Open Orders Value + Total Sell Open Orders Value
		Total Open Buy Orders Value	The value of open buy orders.
		Total Open Sell Orders Value	The value of open sell orders.
Risk Value Controls	Total Risk Value		Total Buy Risk Value + Total Sell Risk Value
	Total Net Risk Value		Total Buy Risk Value – Total Sell Risk Value (absolute value)
		Total Buy Risk Value	Total Buy Traded Value + Total Buy Open Orders Value
		Total Sell Risk Value	Total Sell Traded Value + Total Sell Open Orders Value
Order Count Controls	Total Daily Number of Orders		The number of orders entered during a single trading session.

Order Value is calculated according to formulas indicated below and varies by type of instrument.

Instrument type	Formula for calculating order value
Equities, ETFs and Structured Products	$\text{Price}^1 * \text{Volume}^2$
Bonds	$\text{Nominal} * \text{Volume}^2$
Futures	$\text{Price}^1 * \text{Multiplier} * \text{Volume}^2$
Options	$\text{Multiplier} * \text{Volume}^2 * \text{Strike Price}$

Notation:

- For non-priced buy and sell orders (Market Orders and Market to Limit Orders), the Upper Static Trade Price Collar is used as the Order Price.
 - If Static Trade Price Collar is not activated then Upper Static Order Price Collar is used.
 - If Static Order Price Collar is not activated then Upper Dynamic Trade Price Collar is used.
 - If Dynamic Trade Price Collar is not activated then Upper Dynamic Order Price Collar is used.
 - If no Collars are activated then Last Traded Price is used.
- Volume = Quantity (specified in the order) x Lot Size
- Multiplier for derivatives calculations

Transaction Value is calculated according to the following formulas:

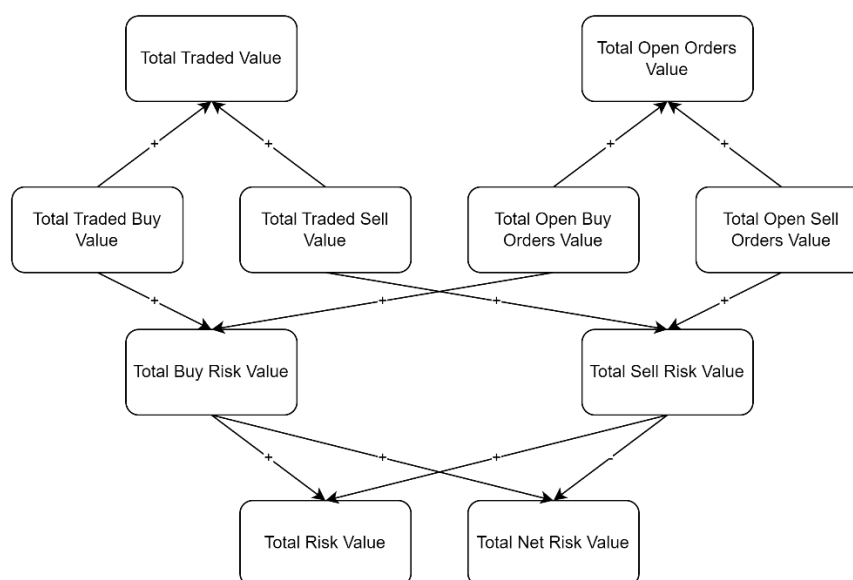
Instrument type	Formula for calculating transaction value
Equities, ETFs and Structured Products	$\text{Price} * \text{Volume}^1$
Bonds	$\text{Nominal} * \text{Volume}^1$
Futures	$\text{Price} * \text{Multiplier} * \text{Volume}^1$
Options	$\text{Multiplier} * \text{Volume}^2 * \text{Strike Price}$

Notation:

- Volume = Quantity (specified in the order) x Lot Size
- Multiplier for derivatives calculations

The mind map illustrates the inter-relationship between the various Post-Trade measures.

Figure 4 - Relationships between Post Trade risk measures



6.2. TRADED VALUE

Traded value control tracks the total value of transactions made by a participant. These controls are computed separately for the buy and sell sides. The table below presents the Traded Value controls implemented within GPW WATS RM.

Category	Risk Measure	Trading Side Risk Measure	Definition	Breach Action
Traded Value Controls	Total Traded Value		Total Traded Buy Value + Total Traded Sell Value	- New Order Entry Rejected - Open Orders Canceled
		Total Traded Buy Value	The value of buy transactions during a trading day.	- New Order Entry Rejected on Buy Side - Open Buy Orders Canceled
		Total Traded Sell Value	The value of sell transactions during a trading day.	- New Order Entry Rejected on Sell Side - Open Sell Orders Canceled

Control Type: Post-trade check

Market Applicability: ~~All Financial Instruments~~ Equities, Bonds, Derivatives

Breach Action: Control triggered when total exceeds the preset limit. Upon breach of limit several action scenarios are available. If the limit is adjusted to a higher amount, the participant will be able to resume trading.
Scope of limit: The limit can be set at the level of Market (MIC code), or market segment

Default limit: none

System level limit: none

Calculation: Transaction value totals are calculated over the trading session for each trading side and then summed up to obtain Total Traded Value.

Category	Risk Metric	Methodology
Traded Value Controls	Total Traded Value	Total Traded Buy Value + Total Traded Sell Value
	Total Traded Buy Value	Running sum of the value of all buy transactions performed during the trading session.
	Total Traded Sell Value	Running sum of the value of all sell transactions performed during the trading session.

Separate value limits may be applied to buy and sell transactions.

Configuration: Three limits are configured for each participant: TotalTradedBuyValueLimit, TotalTradedSellValueLimit and TotalTradedValueLimit.

Design: Part of standard order routing.

6.3. OPEN ORDERS VALUE

Order value control tracks the value of open orders placed by a participant. These controls are computed separately for the buy and sell sides. The table below presents the Order Value controls implemented within GPW WATS RM.

Category	Risk Measure	Trading Side Risk Measure	Definition	Breach Action
Traded Value Controls	Total Open Orders Value		Total Open Buy Orders Value + Total Open Sell Orders Value	New Order Entry Rejected
		Total Open Buy Orders Value	The value of open buy orders during a trading day.	New Order Entry Rejected on Buy Side
		Total Open Sell Orders Value	The value of open sell orders during a trading day.	New Order Entry Rejected on Sell Side

Control Type: Post-trade check

Market Applicability: ~~All Financial Instruments~~ Equities, Bonds, Derivatives

Breach Action: Control triggered when the total exceeds the preset limit. Upon breach of the limit several action scenarios are available and are described in Section 7.1. When the limit is adjusted to a higher level, the participant will be able to resume trading.

Order types: Market, Limit, Iceberg, Stop and modifications.

Scope of limit: The limit can be set at the level of Market (MIC code), or market segment.

Default limit: none

System level limit: none

Calculation: Order value totals are calculated over the trading session for each trading side and then calculated to obtain Total Open Orders Value.

Category	Risk Metric	Methodology
Open Orders Value Controls	Total Open Orders Value	Total Open Buy Orders Value + Total Open Sell Orders Value
	Total Open Buy Orders Value	Sum of the value of all open buy orders
	Total Open Sell Orders Value	Sum of the value of all open sell orders made

Separate value limits may be applied to buy and sell orders.

The following methodology is used when calculating the Total Open Buy Orders Value and Total Open Sell Orders Value metrics. The Total Open Orders Value risk measure is calculated using values of these metrics.

Order category	Order Messages	Methodology
Order entry	NewOrderSingle [type 'D'] OrderAdd	Each order entry message increases Total Open Buy Orders Value or Total Open Sell Orders Value
Order modifications	OrderCancelReplaceRequest [type 'G'] OrderModify	Each order entry modification may change Total Open Buy Orders Value or Total Open Sell Orders Value
Order rejections	BusinessMessageReject [type 'j'] Reject	Each order rejection leaves Total Open Buy Orders Value and Total Open Sell Orders Value unchanged
Order cancels	OrderCancelRequest [type 'F'] OrderCancel	Each order cancellation decreases Total Open Buy Orders Value or Total Open Sell Orders Value

Check application:

FIX Order Gateway		
Order type	Behavior	Control Formula
Buy order	For each new NewOrderSingle {type 'D'} message: - check if Total Open Buy Orders Value less than limit - reject otherwise	$\text{Total Open Buy Orders Value} \leftarrow \text{TotalOpenBuyValueLimit}$
Sell order	For each new NewOrderSingle {type 'D'} message: - check if Total Open Sell Orders Value less than limit - reject otherwise	$\text{Total Open Sell Orders Value} \leftarrow \text{TotalOpenSellValueLimit}$
Buy and Sell orders	For each new NewOrderSingle {type 'D'} message: - check if Total Open Orders Value less than limit - reject otherwise	$\text{Total Open Orders Value} \leftarrow \text{TotalOpenValueLimit}$
Buy order modification	For each new OrderCancelReplaceRequest {type 'G'} message: - check if Total Open Buy Orders Value less than limit - reject otherwise	$\text{Total Open Buy Orders Value} \leftarrow \text{TotalOpenBuyValueLimit}$
Sell order modification	For each new OrderCancelReplaceRequest {type 'G'} message: - check if Total Open Sell Orders Value less than limit - reject otherwise	$\text{Total Open Sell Orders Value} \leftarrow \text{TotalOpenSellValueLimit}$
Buy and Sell order modification	For each new OrderCancelReplaceRequest {type 'G'} message: - check if Total Open Orders Value less than limit - reject otherwise	$\text{Total Open Orders Value} \leftarrow \text{TotalOpenValueLimit}$

Order Value is computed using formulas presented in Section 6.1:

Native Order Gateway		
Order type	Behavior	Formula
Buy order	For each new OrderAdd message: check if Total Open Buy Orders Value less than limit reject otherwise	$\text{Total Open Buy Orders Value} \leftarrow \text{TotalOpenBuyValueLimit}$
Sell order	For each new OrderAdd message: check if Total Open Sell Orders Value less than limit reject otherwise	$\text{Total Open Sell Orders Value} \leftarrow \text{TotalOpenSellValueLimit}$
Buy and Sell orders	For each new OrderAdd message: check if Total Open Orders Value less than limit reject otherwise	$\text{Total Open Orders Value} \leftarrow \text{TotalOpenValueLimit}$
Buy order modification	For each new OrderModify message: check if Total Open Buy Orders Value less than limit reject otherwise	$\text{Total Open Buy Orders Value} \leftarrow \text{TotalOpenBuyValueLimit}$

Native Order Gateway		
Sell order modification	For each new OrderModify message: check if Total Open Sell Orders Value less than limit reject otherwise	Total Open Sell Orders Value < TotalOpenSellValueLimit
Buy and Sell order modification	For each new OrderModify message: check if Total Open Orders Value less than limit reject otherwise	Total Open Orders Value < TotalOpenValueLimit
Order Value is computed using formulas presented in Section 6.1:		

Configuration: Three limits are configured for each participant: TotalOpenBuyValueLimit, TotalOpenSellValueLimit and TotalOpenValueLimit.

Design: Part of the standard order routing. Each incoming order is checked against the limit and rejected if it fails the pre-configured limit check.

6.4. RISK VALUE

Traded value control tracks the total value of transactions made by a participant. Their purpose is to manage the overall risk of a market participant. It is measured as the sum of the Traded Value and Open Orders controls. These controls are calculated separately for the buy and sell sides. The table below presents the Traded Value controls implemented within GPW WATS RM.

Category	Risk Measure	Trading Side Risk Measure	Definition	Breach Action
Traded Value Controls	Total Risk Value		Total Buy Risk Value + Total Sell Risk Value	New Order Entry Rejected
	Total Net Risk Value		Total Buy Risk Value - Total Sell Risk Value	New Order Entry Rejected
		Total Buy Risk Value	Total Traded Buy Value + Total Open Buy Orders Value	New Order Entry Rejected on Buy Side
		Total Sell Risk Value	Total Traded Sell Value + Total Open Sell Orders Value	New Order Entry Rejected on Sell Side

Control Type: Post-trade check

Market Applicability: Equities, [Bonds](#), [Derivatives](#)

Breach Action: Control is triggered when the total exceeds the preset limit. Upon breach of the limit several action scenarios are available and are described in Section 7.1. When the limit is adjusted to a higher level, the participant will be able to resume trading.

Order types: Market, Limit, Iceberg, Stop and modifications.

Scope of limit: The limit can be set at the level of Market (MIC code), or market segment.

Default limit: none

System level limit: none

Calculation: Order value totals are calculated over the trading session for each trading side and then calculated to obtain Total Open Orders Value.

Category	Risk Metric	Calculation
Risk Value Controls	Total Risk Value	Total Buy Risk Value + Total Sell Risk Value
	Total Net Risk Value	Total Buy Risk Value – Total Sell Risk Value (absolute value)
	Total Buy Risk Value	The sum of Total Traded Buy Value and Total Open Buy Orders Value during the trading session.
	Total Sell Risk Value	The sum of Total Traded Sell Value and Total Open Sell Orders Value during the trading session.

Separate value limits may be applied to buy and sell orders. Lot size is taken into account when calculating order values.

The following methodology is used to calculate the risk metrics. Note that the Risk Value risk metrics are a function of Traded Value and Open Orders Value metrics described in Sections 6.2 and 6.3.

Order category	Order Messages	Methodology
Order entry	NewOrderSingle [type 'D'] OrderAdd	Each order entry message increases risk metrics
Order modifications	OrderCancelReplaceRequest [type 'G'] OrderModify	Each order entry modification may change risk metrics
Order rejections	BusinessMessageReject [type 'j'] Reject	Each order rejection leaves risk metrics unchanged
Order cancels	OrderCancelRequest [type 'F'] OrderCancel	Each order cancellation decreases risk metrics

Check application:

FIX Order Gateway		
Order type	Behavior	Control Formula
Buy order	For each new NewOrderSingle [type 'D'] message: - check if Total Buy Risk Value less than limit - reject otherwise	$\text{Total Buy Risk Value} \leq \text{TotalBuyRiskValueLimit}$
Sell order	For each new NewOrderSingle [type 'D'] message: - check if Total Sell Risk Value less than limit - reject otherwise	$\text{Total Sell Risk Value} \leq \text{TotalSellRiskValueLimit}$
Buy and Sell orders	For each new NewOrderSingle [type 'D'] message: - check if Total Risk Value less than limit - check if Total Net Risk Value less than limit - reject otherwise	$\text{Total Risk Value} \leq \text{TotalRiskValueLimit}$ $\text{Total Net Risk Value} \leq \text{TotalNetRiskValueLimit}$
Buy order modification	For each new OrderCancelReplaceRequest [type 'G'] message: - check if Total Buy Risk Value less than limit - reject otherwise	$\text{Total Buy Risk Value} \leq \text{TotalBuyRiskValueLimit}$
Sell order modification	For each new OrderCancelReplaceRequest [type 'G'] message: - check if Total Sell Risk Value less than limit - reject otherwise	$\text{Total Sell Risk Value} \leq \text{TotalSellRiskValueLimit}$

FIX Order Gateway		
Buy and Sell order modification	For each new OrderCancelReplaceRequest (type 'G') message: - check if Total Risk Value less than limit - check if Total Net Risk Value less than limit reject otherwise	$\text{Total Risk Value} \leq \text{TotalRiskValueLimit}$ $\text{Total Net Risk Value} \leq \text{TotalNetRiskValueLimit}$
Order Value is computed using formulas presented in Section 6.1:		

Native Order Gateway		
Order type	Behavior	Formula
Buy order	For each new OrderAdd message: check if Total Buy Risk Value less than limit reject otherwise	$\text{Total Buy Risk Value} \leq \text{TotalBuyRiskValueLimit}$
Sell order	For each new OrderAdd message: check if Total Sell Risk Value less than limit reject otherwise	$\text{Total Sell Risk Value} \leq \text{TotalSellRiskValueLimit}$
Buy and Sell orders	For each new OrderAdd message: - check if Total Risk Value less than limit - check if Total Net Risk Value less than limit reject otherwise	$\text{Total Risk Value} \leq \text{TotalRiskValueLimit}$ $\text{Total Net Risk Value} \leq \text{TotalNetRiskValueLimit}$
Buy order modification	For each new OrderModify message: check if Total Buy Risk Value less than limit reject otherwise	$\text{Total Buy Risk Value} \leq \text{TotalBuyRiskValueLimit}$
Sell order modification	For each new OrderModify message: check if Total Sell Risk Value less than limit reject otherwise	$\text{Total Sell Risk Value} \leq \text{TotalSellRiskValueLimit}$
Buy and Sell order modification	For each new OrderModify message: - check if Total Risk Value less than limit - check if Total Net Risk Value less than limit reject otherwise	$\text{Total Risk Value} \leq \text{TotalRiskValueLimit}$ $\text{Total Net Risk Value} \leq \text{TotalNetRiskValueLimit}$
Order Value is computed using formulas presented in Section 6.1:		

Configuration: Four limits are configured for each participant: TotalBuyRiskValueLimit, TotalSellRiskValueLimit, TotalRiskValueLimit and TotalNetRiskValueLimit.

Design: Part of the standard order routing. Each incoming order is checked against the limit and rejected if it fails the pre-configured limit check.

6.5. TOTAL DAILY NUMBER OF ORDERS

In order to control participant engagement, GPW WATS RM monitors the number of orders entered into the system. This control may serve the purpose of limiting the risks related to runaway algorithmic trading systems or client trading system malfunctions.

The following table shows the types of order messages which are accounted for in the total number of orders statistics under the entry protection scheme. The daily number of orders is calculated over the course of the trading session by counting the number of orders entered up to that point in time. The order "counter" is reset each session.

Order category	Order Messages	Description
Order entry	NewOrderSingle [type 'D'] OrderAdd	New order entries are the key measure of user activity. Persistent orders are also accounted for in order entry statistics.
Order modifications	OrderCancelReplaceRequest [type 'G'] OrderModify	Order modifications are part of the order count statistics
Order rejections	Execution Report [type '8'] OrderAddResponse	Order rejections count the number of messages rejected by the system due to rule violations or business issues.
Order cancels	OrderCancelRequest [type 'F'] OrderCancel	Order cancel requests are part of the order count statistics

Control Type: Post-trade check

Market Applicability: Equities, [Bonds](#), [Derivatives](#)

Breach Action: Control is triggered when the total exceeds the preset limit. Upon breach of the limit several action scenarios are available and are described in Section 7.1. When the limit is adjusted to a higher level, the participant will be able to resume trading.

Order types: Market, Limit, Iceberg, Stop and modifications.

Scope of limit: The limit can be set at the level of Top Level Market Segment (MIC code), or market segment.

Default limit: none

System level limit: none

Calculation: The order count is calculated over the trading session for each type of order. Each trading session starts by initializing the order count to the number of orders which linger from the previous session.

Order category	Order Messages	Methodology
Order entry	NewOrderSingle [type 'D'] OrderAdd	Each order entry message increments order count
Order modifications	OrderCancelReplaceRequest [type 'G'] OrderModify	Each order entry modification increments order count
Order rejections	Execution Report [type '8'] OrderAddResponse	Each order entry rejection increments order count
Order cancels	OrderCancelRequest [type 'F'] OrderCancel	Each order cancellation increments order count

Check application:

Behavior	Control Formula
For each new order message: - check if total number of orders entered during session less than limit - reject otherwise	Order count < MaxOrderCount

Configuration: MaxOrderCount limits are configured for each participant.

Design: Part of standard order routing. Each incoming order is checked against the limit and rejected if it fails the pre-configured limit check.

7. GENERAL CONSIDERATIONS

7.1. BREACH ACTION AND NOTIFICATION

Risk managers configure the behavior of breach actions and notifications. Breach behaviors can be limited to notifications or specific actions.

Breach behavior	Note
Notification	Notifications are sent after a limit is breached. They are sent to both the risk manager and the managed entity.
Action	Actions can be manual or automatic. In the case of manual action, only notifications are issued. If automatic action is configured, a A range of actions can be performed automatically ranging from blocking order entry to order cancellation.

The following actions can be taken either manually or automatically by the risk manager:

Action type	Note
New Order Entry Rejected	Order entry blocked for either buy, sell side or both.
New Order Entry Rejected on Buy Side	
New Order Entry Rejected on Sell Side	
Open Orders Canceled	Cancellation of standing, including limit, stop and iceberg orders. Action can be limited to buy and sell side orders or include both.
Open Buy Orders Canceled	
Open Sell Orders Canceled	

7.2. BREACH ACTION MESSAGE FLOW IN PRE-TRADE CONTROLS

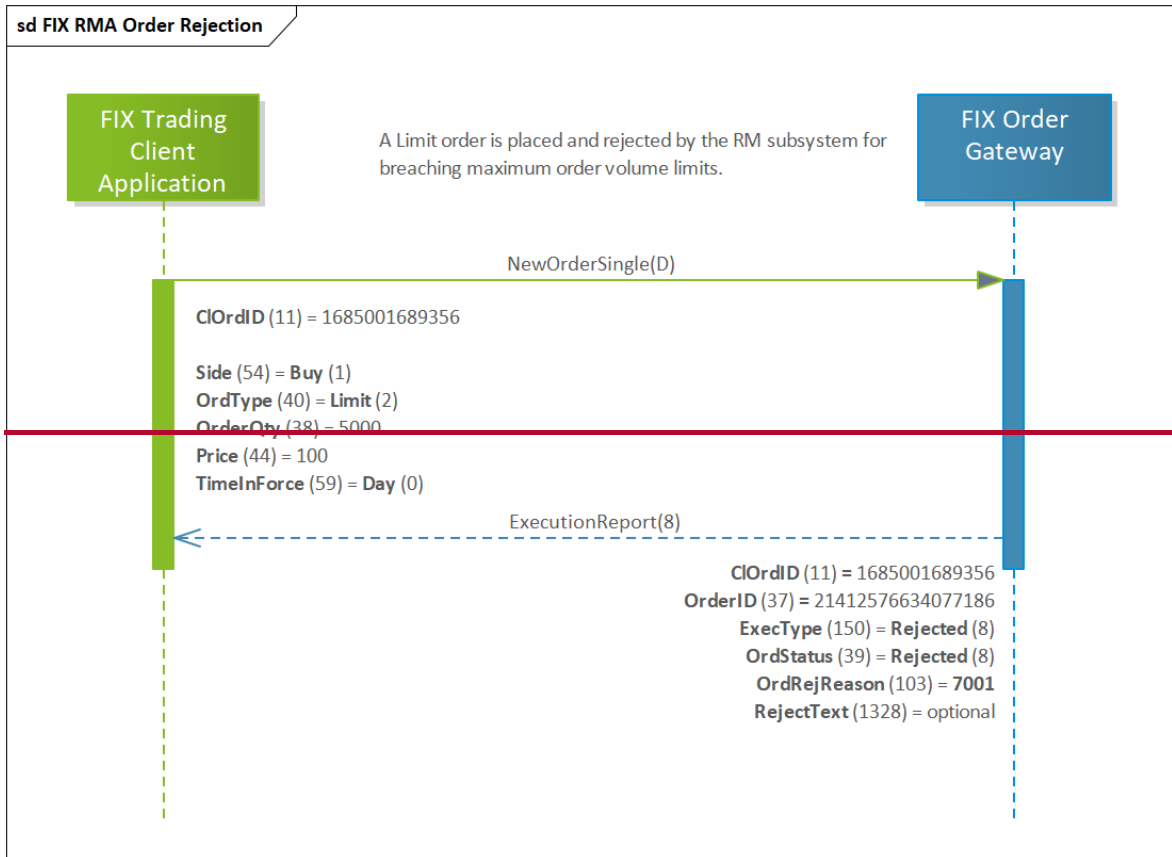
Each time an order or modification breaches a RM limit, a specific sequence of messages is produced by the system. The messages differ by whether order was entered on the Native or FIX trading interface.

The concordance table below presents the set of messages sent out by the system and related parameters in the case of a breach of pre-trade checks (Volume, Value or Price). The scenario assumes that the system received either an OrderAdd/OrderModify on the Native trading port or a NewOrderSingle [D] or OrderCancelReplace [G] on the FIX trading port which breaches a specific pre-trade control.

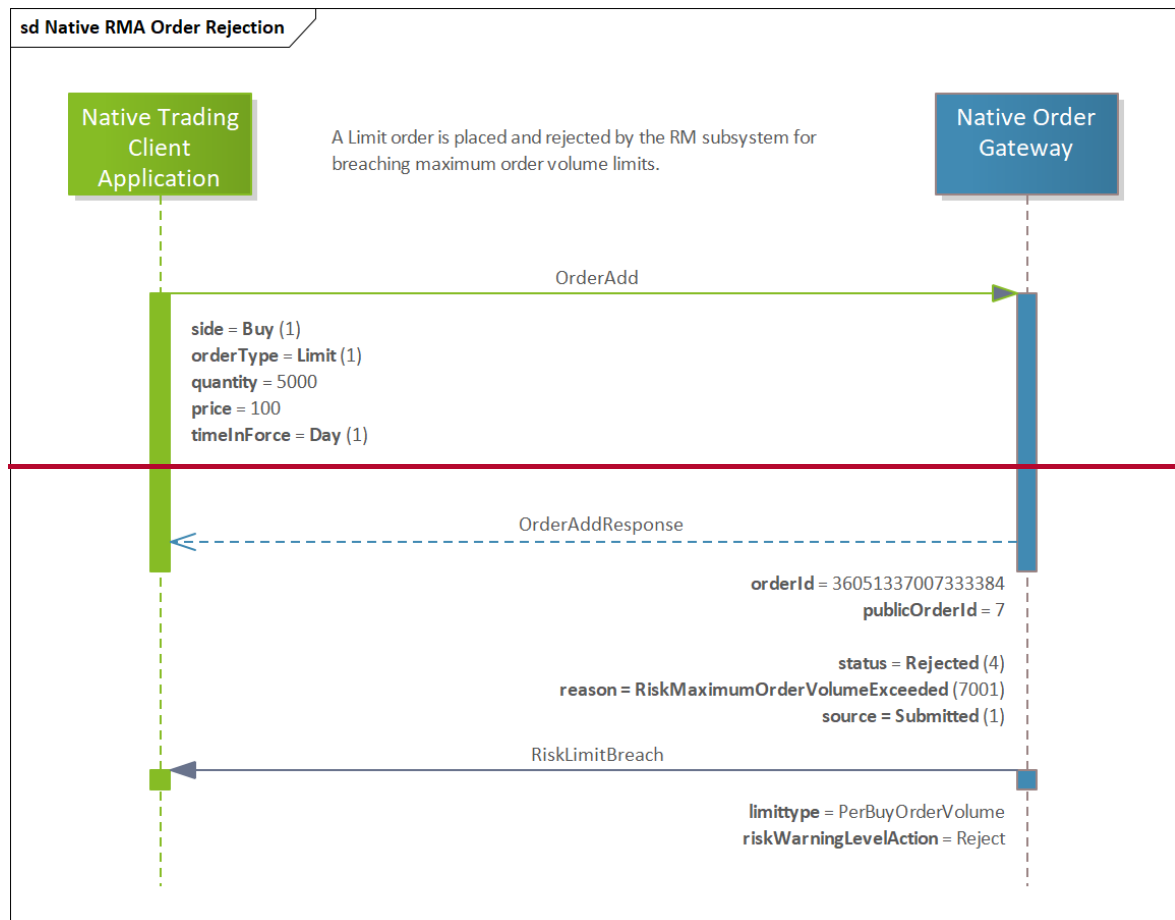
RM Breach Type	Native Trading Port response message		FIX Trading Port response messages
	OrderAddResponse / OrderModifyResponse	RiskLimitBreach	ExecutionReport [8]
Volume	.status=Rejected .reason=7001	.limitType={301-or-311} .riskWarningLevelAction=Reject	OrdStatus [39] = "Rejected" ExecType [150] = "Rejected" OrdRejReason [103] = 7001
Value	.status=Rejected .reason=7002	.limitType={302-or-312} .riskWarningLevelAction=Reject	OrdStatus [39] = "Rejected" ExecType [150] = "Rejected" OrdRejReason [103] = 7002
Price	.status=Rejected .reason=7003	.limitType={313-or-314} .riskWarningLevelAction=Reject	OrdStatus [39] = "Rejected" ExecType [150] = "Rejected"

RM Breach Type	Native Trading Port response message	FIX Trading Port response messages
		OrdRejReason [103] = 7003

7.2.1. SAMPLE KINEMATICS OF FIX PROTOCOL PRE-TRADE CONTROL FAILURE



7.2.2. SAMPLE KINEMATICS OF NATIVE PROTOCOL PRE-TRADE CONTROL FAILURE



7.3. RM PARAMETRIZATION

This section explains the parametrization of the RM system.

We highlight the difference between the concepts of parametrization and configuration of Risk Management limits.

Type of action	Description
Parametrization	Data entered into the parameters of the Risk Management subsystem by Trading Venue staff (the Market Operator) based on contractual agreements between clearing and a trading members.
Configuration	The task of defining the risk limits on pre- and post-trade controls performed by a clearing member for the benefit of a trading member . Risk Controller . Configuration is performed by means of FIX or Native trading port messages such as message PartyRiskLimitsDefinitionRequest (ICS) and RiskLimitDefinition .

7.4. RM CONTROLS CONFIGURATION

The RM system features special errors related to general pre-trade limit definition cases. These are outlined in this section.

7.4.1. LIMITS UNDEFINED / MISSING

Under normal circumstances, the full set of RM pre-trade checks needs to be defined for the Risk Management system to function properly. This means that all 3 pre-trade controls - volume, value, and price - need to be configured.

Failing to define any one of these limits will make the RM system reject any orders entered into the system, even if theoretically, they do not breach a pre-trade check limit. In this case, the error codes listed below will be communicated to the user.

	Native Trading Port response message	FIX Trading Port response messages
	OrderAddResponse / RiskLimitBreach OrderModifyResponse nse	ExecutionReport [8]
	.status=3 Rejected .reason=7000 RiskLimitNotDefined	.limitType=326 .riskWarningLevelAction=Reject Responses.status=3 Rejected Responses.reason=7000 RiskLimitNotDefined
		OrdStatus [39] = "Rejected" ExecType [150] = "Rejected" OrdRejReason [103] = 7000

7.4.2. FEWER THAN 3 PRE-TRADE CONTROLS

GPW WATS can be configured to allow fewer than the full set of 3 pre-trade controls - volume, value, and price - to be defined. If this choice is made, users will be able to define all 3 pre-trade controls, or fewer, or none at all. The consequence of this scheme is that orders entered into the system will be checked only against the pre-trade controls actually defined.

In this case the errors shown in the section above will not display.

This configuration is made in the `all.yaml` file using the `rma_reject_orders_by_missing_prechecks` variable. The configuration scheme is as follows:

- `rma_reject_orders_by_missing_prechecks: true` - definition of the 3 pre-trade controls is mandatory
- `rma_reject_orders_by_missing_prechecks: false` - fewer than 3 pre-trade controls may be defined

By default, the system is configured to require the definition of all 3 pre-trade controls.

7.4.3. CORPORATE CLIENTS

Usually RM pre-trade controls may be defined and applied only to corporate clients. This behavior can be altered through the `all.yaml` file using the `rma_party_role_qualifier_firm_only` variable. The configuration scheme is as follows:

- `rma_party_role_qualifier_firm_only: true` - pre-trade controls may be defined for corporate clients only. If an attempt is made to define a pre-trade check for a non-corporate client, it will be rejected.
- `rma_party_role_qualifier_firm_only: false` - pre-trade controls may be defined for corporate and other clients.

By default, the system is configured to allow the definition of pre-trade controls for corporate clients only.

7.4.4. 7.4.2. FIX AND NATIVE PORT CONFIGURATIONS

The system supports a single RM configuration. The configuration can be performed by commands on the FIX and Native trading ports and orders entered through FIX and Native interfaces undergo identical risk checks.

7.4.5-7.4.3. RM CONFIGURATION ERROR CODES

The process of defining RM ~~pre-trade~~ checks can generate configuration ~~related~~ errors codes. These errors are communicated using ~~Native and~~ FIX Trading Port messages ~~(depending on which port is used to define the RM limits)~~. The following table lists the configuration messages, the configuration response messages and the fields containing the possible error codes.

	Native Trading Port	FIX Trading Port
Pre-trade check configuration message	RiskLimitDefinition	PartyRiskLimitsDefinitionRequest[CS]
Pre-trade check configuration response message	RiskLimitDefinitionResponse reason field	PartyRiskLimitsDefinitionRequestAck[CT] RiskLimitRequestResult [1761]

The table below outlines the error codes which may arise in defining pre-trade checks and their descriptions.

Native Trading Port RiskLimitDefinitionResponse Field: reason			FIX Trading Port PartyRiskLimitsDefinitionRequestAck[CT] Field: RiskLimitRequestResult [1761]		Description
Name			Value		
InvalidParty	1	1	INVALID_PARTY		Invalid party id. Applies when an attempt is made to define a limit on a non-corporate client (rma_party_role_qualifier_firm_only = true)
InvalidRelatedParty	2	2	INVALID_RELATED_PARTY		
InvalidRiskLimitType	3	3	INVALID_RISK_LIMIT_TYPE		Invalid limit type. Corresponds to the types of limits not configurable.
InvalidRiskLimitID	4	4	INVALID_RISK_LIMIT_ID		Unknown limit id. An attempt was made to modify a limit by passing an inexistant limit ID (i.e. manual error).
InvalidRiskLimitAmount	5	5	INVALID_RISK_LIMIT_AMOUNT		Invalid risk limit amount. I.e. a collar multiplier exceeds the allowed range (1%-99910%-99%).
InvalidRiskWarningLevelAction	6	6	INVALID_RISK_WARNING_LEVEL_ACTION		
InvalidRiskInstrumentScope	7	7	INVALID_RISK_INSTRUMENT_SCOPE		Invalid instrument i.e. market segment id or instrument id.
RiskLimitActionsNotSupported	8	8	RISK_LIMIT_ACTIONS_NOT_SUPPORTED		
WarningLevelsNotSupported	9	9	WARNING_LEVELS_NOT_SUPPORTED		
WarningLevelActionsNotSupported	10	10	WARNING_LEVEL_ACTIONS_NOT_SUPPORTED		
RiskInstrumentScopeNotSupported	11	11	RISK_INSTRUMENT_SCOPE_NOT_SUPPORTED		
RiskLimitNotApprovedForParty	12	12	RISK_LIMIT_NOT_APPROVED_FOR_PARTY		
RiskLimitAlreadyDefinedForParty	13	13	RISK_LIMIT_ALREADY_DEFINED_FOR_PARTY		Limit id already defined. Two cases are serviced by this error: 1) user tries to add a risk limit that already exists (duplicate id) 2) user sends an identical limit configuration under a different id - 2 limits with the same content but different IDs are not allowed
InstrumentNotApprovedForParty	14	14	INSTRUMENT_NOT_APPROVED_FOR_PARTY		Invalid instrument i.e. market segment id or instrument id.
NotAuthorized	98	98	NOT_AUTHORIZED		
Other	99	99	OTHER		
NA	100	100	SUCCESSFUL		Not applicable. Used when a configuration was positively validated, and no error is reported.

Native Trading Port RiskLimitDefinitionResponse Field: .reason			FIX Trading Port PartyRiskLimitsDefinitionRequestAck[CT] Field: RiskLimitRequestResult [1761]	Description
MissingMicCode	10 1	10 1	MISSING_MIC_CODE	
InvalidPartyRoleQualifier	10 2	10 2	INVALID_PARTY_ROLE_QUALIFIER	

8. RM PROTOCOL

The RM protocol consists of the following FIX protocol messages.

	Message	Description
FIX	PartyRiskLimitsDefinitionRequest [CS]	Message enabling the definition of a risk limit.
	PartyRiskLimitsDefinitionRequestAck [CT]	Acknowledgment to a PartyRiskLimitsDefinitionRequest [CS] message. Will contain the acknowledgement of a risk limit or the reason for rejecting a risk limit definition.
	PartyRiskLimitsRequest [CL]	Message used to request the state of risk limits configured in the system.
	PartyRiskLimitsReport [CM]	Response to a PartyRiskLimitsRequest [CL] message. It displays the state of the risk limit settings and the utilization of risk limits for a particular user.

8.1. FIX RM PROTOCOL MESSAGES

8.1.1. PARTYRISKLIMITSDEFINITIONREQUEST[CS]

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1666	RiskLimitRequestID	R		Identifier of the Risk Limit request. User generated	string		
	RequestingPartyGrp	R		This group defines the Clearing Member responsible for the risk limits			
1657	NoRequestingPartyIDs	R			NUMINGROUP	1 character	Must be 1 - only one party requests the limit
1658	RequestingPartyID	R		Legal Entity Identifier (LEI)	string		LEI of Clearing Member.
1659	RequestingPartyIDSource	R			char	1 character	N = LEI
1660	RequestingPartyRole	R		Identifies the type or role of the RequestingPartyID(1658) specified.	int	1 character	4 = Clearing Firm
	PartyRiskLimitsUpdateGrp	R		This group defines the number of limits being defined			

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1677	NoPartyRiskLimits	R		Number of limits in the request	NUMINGROUP		Must be 1
1324	ListUpdateAction	R		Risk management action: add, delete or modify a limit	char	1 character	A = Add D = Delete M = Modify
1670	RiskLimitID			Unique reference identifier for a specific risk limit defined for the specified party.	string		
	PartyDetailGrp	R		This group defines the exchange member or client on which the limit is imposed			
1671	NoPartyDetails	R		Number of party details	NUMINGROUP		Must be 1 or 2 The first repeating group defines the exchange member The second repeating group defines the client. If client is not being defined, NoPartyDetails=1
1691	PartyDetailID	R		Identifier of the party	string		LEI of Exchange Member or Short Code of Institutional Client
1692	PartyDetailIDSource	R		Source of the identifier of the PartyDetailID (1691)	char	1 character	P = Short Code N = LEI
1693	PartyDetailRole	R		Identifies the type or role of PartyDetailID (1691)	int		1 = Executing Firm 3 = Client ID
1674	PartyDetailRoleQualifier	C	Conditionally required if PartyDetailedRole(1693) = 2 (Client ID)	Qualifies the value of PartyDetailRole (1693)	int		23 = Firm or Legal Entity
	RiskLimitsGrp	R		This group specifies the limits			
1669	NoRiskLimits			Number of limits defined in the message	NUMINGROUP		Must be 1

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
528	OrderCapacity	O		Designates the capacity of the firm on which the limit is imposed	char	1 character	A = Agency P = Principal
	OrderAttributeGrp	C	Conditionally required if OrderCapacity(528) if filled				
2593	NoOrderAttributes	C		Number of order attribute entries	NUMINGROUP	1 byte	Must be 1
2594	OrderAttributeType	C		The type of order attribute	int		2 = Liquidity Provision Activity Order
2595	OrderAttributeValue	C		The value associated with the order attribute type specified in OrderAttributeType (2594)	string		Y = Yes N = No
	RiskLimitTypesGrp	R		This group specifies the type of limit			
1529	NoRiskLimitTypes	R		The number of risk limit types defined in the message	NUMINGROUP		Must be 1
1530	RiskLimitType	R		The specific limit being defined	int		301 = PER BUY ORDER VOLUME 302 = PER BUY ORDER NOTIONAL VALUE 311 = PER SELL ORDER VOLUME 312 = PER SELL ORDER NOTIONAL VALUE 313 = PER BUY PRICE LIMIT 314 = PER SELL PRICE LIMIT 315 = PER TOTAL BUY TRADED VALUE 316 = PER TOTAL SELL TRADED VALUE 317 = PER TOTAL TRADED VALUE 318 = PER TOTAL BUY OPEN ORDERS VALUE 319 = PER TOTAL SELL OPEN ORDERS VALUE 320 = PER TOTAL OPEN ORDERS VALUE 321 = PER TOTAL BUY RISK VALUE 322 = PER TOTAL SELL RISK VALUE 323 = PER TOTAL RISK VALUE 324 = PER TOTAL NET RISK VALUE 325 = PER TOTAL DAILY NUMBER OF ORDERS 326 = RISK LIMIT NOT DEFINED

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1531	RiskLimitAmount	R		The limit amount to be allocated. Depending on type of limit it is a notional value or quantity.	num	decimal(18,8)	
1767	RiskLimitAction	O		Not applicable	int		
1766	RiskLimitUtilizationAmount	O		Not applicable	AMT		
1765	RiskLimitUtilizationPercent	O		Not applicable	percentage		
	RiskWarningLevelGrp	O		Not applicable			
1559	NoRiskWarningLevels	O		Not applicable	NUMINGROUP		
1769	RiskWarningLevelAction	O		Not applicable	int		
1560	RiskWarningLevelPercent	O		Not applicable	percentage		
1561	RiskWarningLevelName	O		Not applicable	string		
	RiskInstrumentScopeGrp	R		This group defines the instrument (if any) to which the limit is applied			
1534	NoRiskInstrumentScopes	R		The number of instrument scopes	NUMINGROUP		Must be 1
	InstrumentScope	R					
1538	InstrumentScopeSecurityID		Available only for RiskLimitType(1530) = 301; 302; 311; 312; 313; 314	The ID of the security to which limit is applied	string		InstrumentID
1539	InstrumentScopeSecurityIDSource		Available only for RiskLimitType(1530) = 301; 302; 311; 312; 313; 315	The source of security ID (1538)	string		8 = Exchange Symbol
1616	InstrumentScopeSecurityExchange			The trading venue to which limit is applied	string		MIC code
1300	MarketSegmentID			The ID of the market segment to which the limit is applied	string		Market Segment ID

8.1.2. PARTYRISKLIMITSDEFINITIONREQUESTACKICTI

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1666	RiskLimitRequestID	R		Identifier of the Risk Limit request. User generated	string		
1761	RiskLimitRequestResult	<u>C</u>	-Conditionally required if RiskLimitRequestStatus (1762) = 2 (Rejected)	Result of risk limit definition request	int		1 - INVALID_PARTY 2 - INVALID_RELATED_PARTY 3 - INVALID_RISK_LIMIT_TYPE 4 - INVALID_RISK_LIMIT_ID 5 - INVALID_RISK_LIMIT_AMOUNT 6 - INVALID_RISK_WARNING_LEVEL_ACTION 7 - INVALID_RISK_INSTRUMENT_SCOPE 8 - RISK_LIMIT_ACTIONS_NOT_SUPPORTED 9 - WARNING_LEVELS_NOT_SUPPORTED 10 - WARNING_LEVEL_ACTIONS_NOT_SUPPORTED 11 - RISK_INSTRUMENT_SCOPE_NOT_SUPPORTED 12 - RISK_LIMIT_NOT_APPROVED_FOR_PARTY 13 - RISK_LIMIT_ALREADY_DEFINED_FOR_PARTY 14 - INSTRUMENT_NOT_APPROVED_FOR_PARTY 98 - NOT_AUTHORIZED 99 - OTHER 100 - SUCCESSFUL 101 - MISSING_MIC_CODE 102 - INVALID_PARTY_ROLE_QUALIFIER
1762	RiskLimitRequestStatus	R		Status of risk limit definition request	int		0 = Accepted 2 = Rejected
	PartyRiskLimitsAckGrp	R		This group defines the number of limits being defined			
1677	NoPartyRiskLimits	R		Number of limits in the request	NUMINGROUP		Must be 1
1324	ListUpdateAction	R		Risk management action: add, delete or modify a limit	char	1 character	A = Add D = Delete M = Modify

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1670	RiskLimitID			Unique reference identifier for a specific risk limit defined for the specified party.	string		
	PartyDetailGrp	R		This group defines the exchange member or client on which the limit is imposed			
1671	NoPartyDetails	R		Number of party details	NUMINGROUP		Must be 1 or 2 The first repeating group defines the exchange member. The second repeating group defines the client. If client is not being defined, NoPartyDetails=1
1691	PartyDetailID	R		Identifier of the party	string		LEI of Exchange Member or Short Code of Institutional Client
1692	PartyDetailIDSource	R		Source of the identifier of the PartyDetailID (1691)	char	1 character	P = Short Code N = LEI
1693	PartyDetailRole	R		Identifies the type or role of PartyDetailID (1691)	int		1 = Executing Firm 3 = Client ID
1674	PartyDetailRoleQualifier	C	Conditionally required if PartyDetailRole(1693) = 2 (Client ID)	Qualifies the value of PartyDetailRole (1693)	int		23 = Firm or Legal Entity
	RiskLimitsGrp	R		This group specifies the limits			
1669	NoRiskLimits			Number of limits defined in the message	NUMINGROUP		Must be 1
528	OrderCapacity	O		Designates the capacity of the firm on which the limit is imposed	char	1 character	A = Agency P = Principal
	OrderAttributeGrp	C	Conditionally required if OrderCapacity(528) is filled				

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
2593	NoOrderAttributes	C		Number of order attribute entries	NUMINGROUP	1 byte	Must be 1
2594	OrderAttributeType	C		The type of order attribute	int		2 = Liquidity Provision Activity Order
2595	OrderAttributeValue	C		The value associated with the order attribute type specified in OrderAttributeType (2594)	string		Y = Yes N = No
	RiskLimitTypesGrp	R		This group specifies the type of limit			
1529	NoRiskLimitTypes	R		The number of risk limit types defined in the message	NUMINGROUP		Must be 1
1530	RiskLimitType	R		The specific limit being defined	int		301 = PER BUY ORDER VOLUME 302 = PER BUY ORDER NOTIONAL VALUE 311 = PER SELL ORDER VOLUME 312 = PER SELL ORDER NOTIONAL VALUE 313 = PER BUY PRICE LIMIT 314 = PER SELL PRICE LIMIT 315 = PER TOTAL BUY TRADED VALUE 316 = PER TOTAL SELL TRADED VALUE 317 = PER TOTAL TRADED VALUE 318 = PER TOTAL BUY OPEN ORDERS VALUE 319 = PER TOTAL SELL OPEN ORDERS VALUE 320 = PER TOTAL OPEN ORDERS VALUE 321 = PER TOTAL BUY RISK VALUE 322 = PER TOTAL SELL RISK VALUE 323 = PER TOTAL RISK VALUE 324 = PER TOTAL NET RISK VALUE 325 = PER TOTAL DAILY NUMBER OF ORDERS 326 = RISK LIMIT NOT DEFINED
531	RiskLimitAmount	R		The limit amount to be allocated. Depending on type of limit it is a notional value or quantity.	num	decimal(18,8)	
1767	RiskLimitAction	O		Not applicable	int		
1766	RiskLimitUtilizationAmount	O		Not applicable	AMT		

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1765	RiskLimitUtilizationPercent	O		Not applicable	percentage		
	RiskWarningLevelGrp	O		Not applicable			
1559	NoRiskWarningLevels	O		Not applicable	NUMINGROUP		
1769	RiskWarningLevelAction	O		Not applicable	int		
1560	RiskWarningLevelPercent	O		Not applicable	percentage		
1561	RiskWarningLevelName	O		Not applicable	string		
	RiskInstrumentScopeGrp	R		This group defines the instrument (if any) to which the limit is applied			
1534	NoRiskInstrumentScopes	R		The number of instrument scopes	NUMINGROUP		Must be 1
	InstrumentScope	R					
1538	InstrumentScopeSecurityID			The ID of the security to which limit is applied	string		InstrumentID
1539	InstrumentScopeSecurityIDSource			The source of security ID (1538)	string		8 = Exchange Symbol
1616	InstrumentScopeSecurityExchange			The trading venue to which limit is applied	string		MIC code
1300	MarketSegmentID			The ID of the market segment to which the limit is applied	string		Market Segment ID

8.1.3. PARTYRISKLIMITSREQUEST[CL]

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1666	RiskLimitRequestID	R		Unique identifier for the PartyRiskLimitsRequest	string		
1760	RiskLimitRequestType	R		The type of request relative to risk limits.	int		1 = Definitions 3 = Definitions and utilization
	RequestingPartyGrp	R		This group defines the Clearing Member making the request. Responsible for risk limits			
1657	NoRequestingPartyIDs	R			NUMINGROUP	1 character	Must be 1 - only one party requests the limit
1658	RequestingPartyID	R		Legal Entity Identifier (LEI)	string		LEI of Clearing Member.
1659	RequestingPartyIDSource	R			char	1 character	N = LEI
1660	RequestingPartyRole	R		Identifies the type or role of the RequestingPartyID(1658) specified.	int	1 character	4 = Clearing Firm
	Parties						
453	NoPartyIDs	R		Number of PartyIDs.	NUMINGROUP		Must be 1 or 2 The first repeating group defines the exchange member The second repeating group defines the client. If client is not being defined, NoPartyDetails=1
448	PartyID	R		Identifier of the party	string		LEI of Exchange Member or Short Code of Institutional Client
447	PartyIDSource	R		Identifies class or source of the PartyID (448)	char	1 character	P = Short Code N = LEI
452	PartyRole	R		Identifies the type or role of the PartyID (448).	int		1 = Executing Firm 3 = Client ID
2376	PartyRoleQualifier	C	Conditionally required if PartyRole (452) = 23 (Client ID)	Qualifies the value of PartyRole(452).	int		23 = Firm or Legal Entity
	RequestedRiskLimitTypesGrp	O					
1668	NoRequestedRiskLimitType	O		Number of risk limit types requested.	NUMINGROUP		

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
528	OrderCapacity	O		Designates the capacity of the firm on which the limit is imposed	char	1 character	A = Agency P = Principal
	OrderAttributeGrp	O	Conditionally required if OrderCapacity(528) if filled				
2593	NoOrderAttributes	O		Number of order attribute entries	NUMINGROUP	1 byte	Must be 1
2594	OrderAttributeType	O		The type of order attribute	int		2 = Liquidity Provision Activity Order
2595	OrderAttributeValue	O		The value associated with the order attribute type specified in OrderAttributeType (2594)	string		Y = Yes N = No
1530	RiskLimitType	O		Used to specify the type of risk limit amount or position limit quantity or margin requirement amounts.	int		301 = PER BUY ORDER VOLUME 302 = PER BUY ORDER NOTIONAL VALUE 311 = PER SELL ORDER VOLUME 312 = PER SELL ORDER NOTIONAL VALUE 313 = PER BUY PRICE LIMIT 314 = PER SELL PRICE LIMIT 315 = PER TOTAL BUY TRADED VALUE 316 = PER TOTAL SELL TRADED VALUE 317 = PER TOTAL TRADED VALUE 318 = PER TOTAL BUY OPEN ORDERS VALUE 319 = PER TOTAL SELL OPEN ORDERS VALUE 320 = PER TOTAL OPEN ORDERS VALUE 321 = PER TOTAL BUY RISK VALUE 322 = PER TOTAL SELL RISK VALUE 323 = PER TOTAL RISK VALUE 324 = PER TOTAL NET RISK VALUE 325 = PER TOTAL DAILY NUMBER OF ORDERS 326 = RISK LIMIT NOT DEFINED
	RiskInstrumentScopeGrp	O		This group defines the instrument (if any) to which the limit is applied			
1534	NoRiskInstrumentScopes	O		The number of instrument scopes	NUMINGROUP		Must be 1

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
	InstrumentScope	O					
1538	InstrumentScopeSecurityID	O	Available only for RiskLimitType (1530) = 301; 302; 311; 312; 313; 314	The ID of the security to which limit is applied	string		InstrumentID
1539	InstrumentScopeSecurityIDSource	O	Available only for RiskLimitType (1530) = 301; 302; 311; 312; 313; 315	The source of security ID (1538)	string		8 = Exchange Symbol
1616	InstrumentScopeSecurityExchange	O		The trading venue to which limit is applied	string		MIC code
1300	MarketSegmentID	O		The ID of the market segment to which the limit is applied	string		Market Segment ID

8.1.4. PARTYRISKLIMITSREPORT[CM]

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1667	RiskLimitReportID	R		Identifier for the PartyRiskLimitsReport	string		
1666	RiskLimitRequestID	C	Conditionally required when responding to Party Risk Limits Request	Unique identifier for the PartyRiskLimitsRequest	string		
1760	RiskLimitRequestType	C	Conditionally required when responding to Party Risk Limits Request	Type of risk limit information	int		1 = Definitions 3 = Definitions and utilization
1511	RequestResult	C	Conditionally required when responding to Party Risk Limits Request	Result of a request as identified by the appropriate request ID field	int		0 = Valid request 1 = Invalid or unsupported request 2 = No data found that match selection criteria 3 = Not authorized to retrieve data 4 = Data temporarily unavailable 5 = Request for data not supported 99 = Other
325	UnsolicitedIndicator	R		Indicates whether or not message is being sent as a result of a subscription request or not	Boolean		N = Message is being sent as a result of a prior request Y = Message is being sent unsolicited
	PartyRiskLimitsGrp	R		This group defines the number of limits being defined			
1677	NoPartyRiskLimits	R		Number of party risk limits	NUMINGROUP		
	PartyDetailGrp	R		This group defines the exchange member or client on which the limit is imposed			

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1671	NoPartyDetails	R		Number of party details	NUMINGROUP		Must be 1 or 2 The first repeating group defines the exchange member The second repeating group defines the client. If client is not being defined, NoPartyDetails=1
1691	PartyDetailID	R		Identifier of the party	string		LEI of Exchange Member Or Short Code of Institutional Client
1692	PartyDetailIDSource	R		Source of the identifier of the PartyDetailID (1691)	char	1 character	P = Short Code N = LEI
1693	PartyDetailRole	R		Identifies the type or role of PartyDetailID (1691)	int		1 = Executing Firm 3 = Client ID
1674	PartyDetailRoleQualifier	C	Conditionally required if PartyDetailedRole(1693) = 2 (Client ID)	Qualifies the value of PartyDetailRole (1693)	int		23 = Firm or Legal Entity
	RiskLimitsGrp	R		This group specifies the limits			
1669	NoRiskLimits			Number of limits defined in the message	NUMINGROUP		Must be 1
528	OrderCapacity	O		Designates the capacity of the firm on which the limit is imposed	char	1 character	A = Agency P = Principal
	OrderAttributeGrp	C	Conditionally required if OrderCapacity(528) if filled				
2593	NoOrderAttributes	C		Number of order attribute entries	NUMINGROUP	1 byte	Must be 1
2594	OrderAttributeType	C		The type of order attribute	int		2 = Liquidity Provision Activity Order
2595	OrderAttributeValue	C		The value associated with the order attribute type specified in OrderAttributeType (2594)	string		Y = Yes N = No
	RiskLimitTypesGrp	R		This group specifies the type of limit			
1529	NoRiskLimitTypes	R		The number of risk limit types defined in the message	NUMINGROUP		Must be 1

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1530	RiskLimitType	R		The specific limit being defined	int		301 = PER BUY ORDER VOLUME 302 = PER BUY ORDER NOTIONAL VALUE 311 = PER SELL ORDER VOLUME 312 = PER SELL ORDER NOTIONAL VALUE 313 = PER BUY PRICE LIMIT 314 = PER SELL PRICE LIMIT 315 = PER TOTAL BUY TRADED VALUE 316 = PER TOTAL SELL TRADED VALUE 317 = PER TOTAL TRADED VALUE 318 = PER TOTAL BUY OPEN ORDERS VALUE 319 = PER TOTAL SELL OPEN ORDERS VALUE 320 = PER TOTAL OPEN ORDERS VALUE 321 = PER TOTAL BUY RISK VALUE 322 = PER TOTAL SELL RISK VALUE 323 = PER TOTAL RISK VALUE 324 = PER TOTAL NET RISK VALUE 325 = PER TOTAL DAILY NUMBER OF ORDERS 326 = RISK LIMIT NOT DEFINED
1531	RiskLimitAmount	R		The limit amount to be allocated. Depending on type of limit it is a notional value or quantity.	num	decimal(18,8)	
1767	RiskLimitAction	C	Conditionally required if UnsolicitedIndicator (325) = Y (Message is being sent unsolicited)	The action to be taken in the case the limit is breached	int		4 = Warning 9 = Suspend

Tag	Field Name	Required	Conditional	Description	Data Type	Max Length	Values
1766	RiskLimitUtilizationAmount	C	Conditionally required if RiskLimitRequestType (1760) = 3 (Definitions and utilization)	Risk Limit Utilization Amount	AMT		
1765	RiskLimitUtilizationPercent	C	Conditionally required if: • RiskLimitRequestType (1760) = 3 (Definitions and utilization) • Or • UnsolicitedIndicator (325) = Y (Message is being sent unsolicited)	Risk Limit Utilization Percent	percentage		
	RiskWarningLevelGrp	C					
1559	NoRiskWarningLevels	C	Conditionally required if RiskLimitAction (1767) = 4 (Warning)	Number of risk warning levels	NUMINGROUP		Value can only be 1.
1769	RiskWarningLevelAction	C	Conditionally required if RiskLimitAction (1767) = 4 (Warning)	Action to take should warning level be exceeded	int		4 = Warning
1560	RiskWarningLevelPercent	C	Conditionally required if RiskLimitAction (1767) = 4 (Warning)		percentage		
1561	RiskWarningLevelName	C	Conditionally required if RiskLimitAction (1767) = 4 (Warning)	Name associated with the risk warning level	string		Valid values: 1 = Warning Level 1 Breach 2 = Warning Level 2 Breach
	RiskInstrumentScopeGrp	R		This group defines the instrument (if any) to which the limit is applied			
1534	NoRiskInstrumentScopes	R		The number of instrument scopes	NUMINGROUP		Must be 1
	InstrumentScope	R					
1538	InstrumentScopeSecurityID			The ID of the security to which limit is applied	string		InstrumentID
1539	InstrumentScopeSecurityIDSource			The source of security ID (1538)	string		8 = Exchange Symbol
1616	InstrumentScopeSecurityExchange			The trading venue to which limit is applied	string		MIC code
1300	MarketSegmentID			The ID of the market segment to which the limit is applied	string		Market Segment ID

8.2. FIX RM PROTOCOL MESSAGE EXAMPLES

8.2.1. PARTYRISKLIMITSDEFINITIONREQUEST[CS]

This section outlines sample messages for configuring risk limits for screening:

- agency and principal orders
- institutional orders

8.2.1.1. Agency and Principal orders

Tag	Field Name	Required	All orders	All agency orders (non market making)	All agency orders (market making) all external market makers	All principal orders (non market making)	All principal orders (market making)
1666	RiskLimitRequestID	R	1	2	3	4	5
	RequestingPartyGrp	R					
1657	NoRequestingPartyIDs	R	1	1	1	1	1
1658	RequestingPartyID	R	LEI #	LEI #	LEI #	LEI #	LEI #
1659	RequestingPartyIDSource	R	N	N	N	N	N
1660	RequestingPartyRole	R	4	4	4	4	4
	PartyRiskLimitsUpdateGrp	R					
1677	NoPartyRiskLimits	R	1	1	1	1	1
1324	ListUpdateAction	R	A	A	A	A	A
1670	RiskLimitID		100	100	101	102	103
	PartyDetailGrp	R					
1671	NoPartyDetails	R	1	1	1	1	1
1691	PartyDetailID	R	LEI #	LEI #	LEI #	LEI #	LEI #
1692	PartyDetailIDSource	R	N	N	N	N	N

Tag	Field Name	Required	All orders	All agency orders (non market making)	All agency orders (market making) all external market makers	All principal orders (non market making)	All principal orders (market making)
1693	PartyDetailRole	R	1	1	1	1	1
1674	PartyDetailRoleQualifier	C Conditionally required if PartyDetailedRole(1693) = 2 (Client ID)	none	none	none	none	none
	RiskLimitsGrp	R					
1669	NoRiskLimits		none	1	1	1	1
528	OrderCapacity	O	none	A	A	P	P
	OrderAttributeGrp	C Conditionally required if OrderCapacity(528) if filled					
2593	NoOrderAttributes	C	none	1	1	1	1
2594	OrderAttributeType	C	none	2	2	2	2
2595	OrderAttributeValue	C	none	N	Y	N	Y
	RiskLimitTypesGrp	R					
1529	NoRiskLimitTypes	R		1		1	1
1530	RiskLimitType	R		301		302	302
1531	RiskLimitAmount	R		1,000,000	1,000,000	100,000,000	100,000,000
1767	RiskLimitAction	O					
1766	RiskLimitUtilizationAmount	O					
1765	RiskLimitUtilizationPercent	O					
	RiskWarningLevelGrp	O					

Tag	Field Name	Required	All orders	All agency orders (non market making)	All agency orders (market making) all external market makers	All principal orders (non market making)	All principal orders (market making)
9	155	NoRiskWarningLevels	O				
9	176	RiskWarningLevelAction	O				
0	156	RiskWarningLevelPercent	O				
1	156	RiskWarningLevelName	O				
		RiskInstrumentScopeGrp	R				
1534		NoRiskInstrumentScopes	R	1	1	1	1
		InstrumentScope	R				
8	153	InstrumentScopeSecurityID	Available only for RiskLimitType(1530) = 301; 302; 311; 312; 313; 314				
9	153	InstrumentScopeSecurityIDSource	Available only for RiskLimitType(1530) = 301; 302; 311; 312; 313; 315				
6	161	InstrumentScopeSecurityExchange		XWAR	XWAR		
0	130	MarketSegmentID				100	100

8.2.1.2. Institutional orders

Tag	Field Name	Required	Institutional client orders (non-market making) incl. sponsor access client		Institutional client orders (market making) external market maker incl. sponsor access client	
1666	RiskLimitRequestID	R	6		7	
	RequestingPartyGrp	R				
1657	NoRequestingPartyIDs	R	1		1	
1658	RequestingPartyID	R	LEI 999		LEI 999	
1659	RequestingPartyIDSource	R	N		N	
1660	RequestingPartyRole	R	4		4	
	PartyRiskLimitsUpdateGrp	R				
1677	NoPartyRiskLimits	R	1		1	
1324	ListUpdateAction	R	A		A	
1670	RiskLimitID		104		105	
	PartyDetailGrp	R				
1671	NoPartyDetails	R	2		2	
			1 st repeating group	2 nd repeating group	1 st repeating group	2 nd repeating group
1691	PartyDetailID	R	LEI 888	100	LEI 888	200
1692	PartyDetailIDSource	R	N	P	N	P
1693	PartyDetailRole	R	1	3	1	3
1674	PartyDetailRoleQualifier	C Conditionally required if PartyDetailedRole(1693) = 2 (Client ID)	none	23	none	23
	RiskLimitsGrp	R				
1669	NoRiskLimits		1		1	
528	OrderCapacity	O	A		A	
	OrderAttributeGrp	C Conditionally required if OrderCapacity(528) if filled				
2593	NoOrderAttributes	C	1		1	

Tag	Field Name	Required	Institutional client orders (non-market making) incl. sponsor access client		Institutional client orders (market making) external market maker incl. sponsor access client	
2594	OrderAttributeType	C	2		2	
2595	OrderAttributeValue	C	N		Y	
	RiskLimitTypesGrp	R				
1529	NoRiskLimitTypes	R	1		1	
1530	RiskLimitType	R	302		302	
1531	RiskLimitAmount	R	100,000,000		100,000,000	
1767	RiskLimitAction	O				
1766	RiskLimitUtilizationAmount	O				
1765	RiskLimitUtilizationPercent	O				
	RiskWarningLevelGrp	O				
559	NoRiskWarningLevels	O				
769	RiskWarningLevelAction	O				
560	RiskWarningLevelPercent	O				
561	RiskWarningLevelName	O				
	RiskInstrumentScopeGrp	R				
1534	NoRiskInstrumentScopes	R	1		1	
	InstrumentScope	R				
538	InstrumentScopeSecurityID	Available only for RiskLimitType(1530) = 301; 302; 311; 312; 313; 314				
539	InstrumentScopeSecurityIDSource	Available only for RiskLimitType(1530) = 301; 302; 311; 312; 313; 315				
616	InstrumentScopeSecurityExchange					
300	MarketSegmentID		100		100	

8.2.2. PARTYRISKLIMITSREPORT[CM]

The table below shows examples of response messages,

- in response to CL limit requests
- unsolicited messages sent in cases where either a warning level or a limit is breached.

Tag	Field Name	Required	Conditional	Responses to CL requests		Unsolicited messages	
				Extent of limits	Extent of limits and utilization	Information about exceeding a warning level	Information about exceeding a limit
1667	RiskLimitReportID	R					
1666	RiskLimitRequestID	C	Conditionally required when responding to Party Risk Limits Request				
1760	RiskLimitRequestType	C	Conditionally required when responding to Party Risk Limits Request				
1511	RequestResult	C	Conditionally required when responding to Party Risk Limits Request				
325	UnsolicitedIndicator	R					
	PartyRiskLimitsGrp	R					
1677	NoPartyRiskLimits	R					
	PartyDetailGrp	R					
1671	NoPartyDetails	R					
1691	PartyDetailID	R					
1692	PartyDetailIDSource	R					
1693	PartyDetailRole	R					

Tag	Field Name	Required	Conditional	Responses to CL requests		Unsolicited messages	
1674	PartyDetailRoleQualifier	C	Conditionally required if PartyDetailedRole(1693) = 2 (Client ID)				
	RiskLimitsGrp	R					
1669	NoRiskLimits						
528	OrderCapacity	O					
	OrderAttributeGrp	C	Conditionally required if OrderCapacity(528) if filled				
2593	NoOrderAttributes	C					
2594	OrderAttributeType	C					
2595	OrderAttributeValue	C					
	RiskLimitTypesGrp	R					
1529	NoRiskLimitTypes	R		1	1	1	1
1530	RiskLimitType	R		301	3015	3015	3015
1531	RiskLimitAmount	R		100,000	100,000	100,000	100,000
1767	RiskLimitAction	C	Conditionally required if UnsolicitedIndicator (325) = Y (Message is being sent unsolicited)			4	9
1766	RiskLimitUtilizationAmount	C	Conditionally required if RiskLimitRequestType (1760) = 3 (Definitions and utilization)		52,000	52,000	109,000
1765	RiskLimitUtilizationPercent	C	Conditionally required if: - RiskLimitRequestType (1760) = 3 (Definitions and utilization) or - UnsolicitedIndicator (325) = Y (Message is being sent unsolicited)		52%	52%	109%
	RiskWarningLevelGrp	C					

Tag	Field Name	Required	Conditional	Responses to CL requests		Unsolicited messages	
1559	NoRiskWarningLevels	C	Conditionally required if RiskLimitAction (1767) = 4 (Warning)			1	
1769	RiskWarningLevelAction	C	Conditionally required if RiskLimitAction (1767) = 4 (Warning)			4	
1560	RiskWarningLevelPercent	C	Conditionally required if RiskLimitAction (1767) = 4 (Warning)			50%	
1561	RiskWarningLevelName	C	Conditionally required if RiskLimitAction (1767) = 4 (Warning)			1	
	RiskInstrumentScopeGrp	R					
1534	NoRiskInstrumentScopes	R					
	InstrumentScope	R					
1538	InstrumentScopeSecurityID						
1539	InstrumentScopeSecurityIDSource						
1616	InstrumentScopeSecurityExchange						
130	MarketSegmentID						

8.3. RM CONTROL OF TRADING ACTIVITY

The Risk Management subsystem controls order entry messages submitted on the FIX and Native trading ports. The following order messages are controlled by the Risk Management subsystem viz pre and post trade checks:

8.3. MESSAGE KINEMATICS

8.3.1. DEFINING A RISK LIMIT

This example shows the kinematics of defining a Maximum Buy Order Volume limit. The kinematics show a PartyRiskLimitsDefinitionRequestAck (CT) Message being returned in response to a PartyRiskLimitsDefinitionRequest (CS) message.

1. The limit is defined by a clearing firm | LEIcode (FIXTag: 1658) = 1112222ClearingFirm.
2. The limit applies to:
 - a) orders submitted by Exchange Member | LEI (FIXTag: 1691) = 8889999EXECUTINGFIRM.
 - b) which concern clients | OrderCapacity (FIXTag: 528) = AGENCY (A).
 - c) who do not act as market makers | OrderAttribute (FIXTag: 2594) = Liquidity Provision Activity Order (2) and OrderAttributeValue (FIXTag: 2595) = No (N).
 - d) apply to shares listed on the Regulated Market | MICCode (FIXTag: 1616) = XWAR.
3. The value of the limit is set to 200,000 currency units | RiskLimitAmount (FIXTag: 1531) = 200000.

FIX Trading Client Risk Management Application

WATS FIX Trading Port

PartyRiskLimitsDefinitionRequest(CS)

RiskLimitRequestID (1666) = 100

RequestingPartyGrp

NoRequestingPartyIDs (1657) = 1
RequestingPartyID (1658) = 11112222CLEARINGFIRM
RequestingPartyIDSource (1659) = LEI (N)
RequestingPartyRole (1660) = Clearing Firm (4)

PartyRiskLimitsUpdateGrp

NoPartyRiskLimits (1677) = 1
ListUpdateAction (1324) = Add (A)
RiskLimitID (1670) = 123

PartyDetailGrp

NoPartyDetails (1671) = 1
PartyDetailID (1691) = 8889999EXECUTINGFIRM
PartyDetailIDSource (1692) = LEI (N)
PartyDetailRole (1693) = Executing Firm (1)
PartyDetailRoleQualifier (1674) = empty

RiskLimitsGrp

NoRiskLimits (1669) = 1
OrderCapacity (528) = Agency (A)

OrderAttributeGrp

NoOrderAttributes (2593) = 1
OrderAttributeType (2594) = Liquidity Provision Activity Order (2)
OrderAttributeValue (2595) = No (N)

RiskLimitTypesGrp

NoRiskLimitTypes (1592) = 1
RiskLimitType (1530) = PER BUY ORDER VOLUME (301)
RiskLimitAmount (1531) = 200000
[...]

RiskInstrumentScopeGrp

NoRiskInstrumentScopes (1534) = 1

InstrumentScopeGrp

InstrumentScopeSecurityID (1538) = Empty
InstrumentScopeSecurityIDSource (1539) = Empty
MarketSegmentID (1300) = Empty
InstrumentScopeSecurityExchange (1616) = XWAR

PartyRiskLimitsDefinitionRequestAck(CT)

RiskLimitRequestID (1666) = 1
RiskLimitRequestResult (1761) = Empty
RiskLimitRequestStatus (1762) = Accepted (0)

PartyRiskLimitsAckGrp

NoPartyRiskLimits (1677) = 1
ListUpdateAction (1324) = Add (A)
RiskLimitID (1670) = 123

[...]

8.3.1.1. Other cases of configuring risk limits across an Exchange Member's activity

BIN	All agency orders Non market maker		All agency orders Market maker		All principal orders Non market maker		All principal orders Market maker	
RiskLimitsGrp								
NoRiskLimits(1669)	1		1		1		1	
OrdeCapacity(528)	A	A	P	P				
OrderAttributeGrp								
NoOrderAttributes(2593)	1	1	1	1				
OrderAttributeType(2594)	2	2	2	2				
OrderAttributeValue(2595)	N	Y	N	Y				

8.3.1.2. Example of configuring a risk limit for an institutional client

The client does not act as market maker and is identified by Short Code = 100.

	Value
PartyDetailGrp	
NoPartyDetails(1671)	2
<i>First repeating group: exchange member</i>	
PartyDetailID(1691)	8889999EXECUTINGFIRM
PartyDetailIDSource(1692)	LEI (N)
PartyDetailRole(1693)	Executing Firm (1)
PartyDetailRoleQualifier(1694)	Empty
<i>Second repeating group: institutional client</i>	
PartyDetailID(1691)	100
PartyDetailIDSource(1692)	Short Code (P)
PartyDetailRole(1693)	ClientID (3)
PartyDetailRoleQualifier(1694)	Firm or Legal Entity (23)
RiskLimitsGrp	
NoRiskLimits(1669)	1
OrdeCapacity(528)	A
OrderAttributeGrp	
NoOrderAttributes(2593)	1
OrderAttributeType(2594)	2
OrderAttributeValue(2595)	N

8.3.1.3. Other cases of configuring risk limits across the market structure

Example 1. Risk limit configured at the level of an individual financial instrument using InstrumentID

	Value
<u>RiskInstrumentScopeGroup</u>	
<u>NoRiskInstrumentScopes(1534)</u>	<u>1</u>
<u>InstrumentScopeGrp</u>	
<u>InstrumentScopeSecurityID(1538)</u>	<u>1598</u>
<u>InstrumentScopeSecurityIDSource(1539)</u>	<u>Exchange Symbol (8)</u>
<u>MarketSegmentID(1300)</u>	<u>Empty</u>
<u>InstrumentScopeSecurityExchange(1616)</u>	<u>Empty</u>

Example 2. Risk limit configured at the level of a market segment using SegmentID

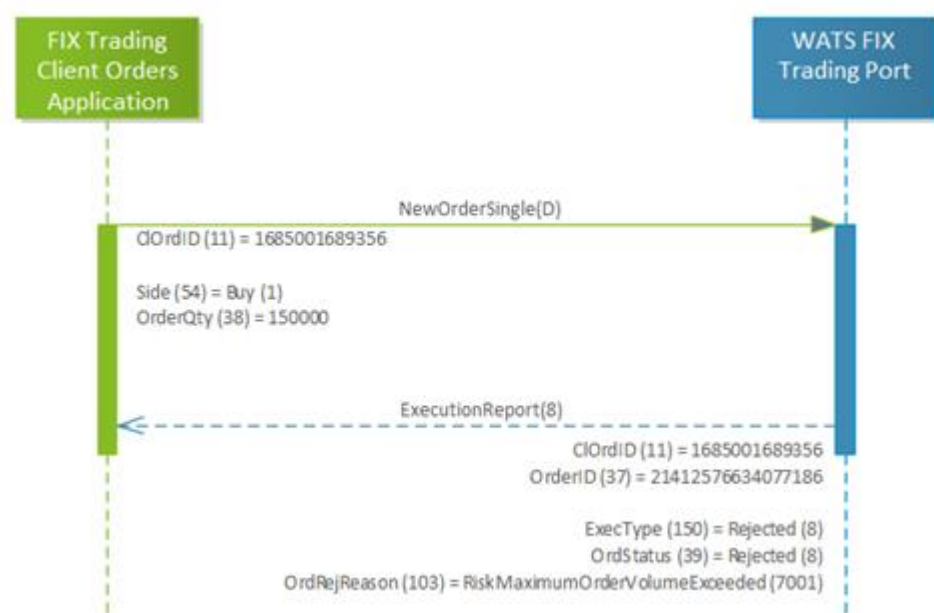
	Value
<u>RiskInstrumentScopeGroup</u>	
<u>NoRiskInstrumentScopes(1534)</u>	<u>1</u>
<u>InstrumentScopeGrp</u>	
<u>InstrumentScopeSecurityID(1538)</u>	<u>Empty</u>
<u>InstrumentScopeSecurityIDSource(1539)</u>	<u>Empty</u>
<u>MarketSegmentID(1300)</u>	<u>201</u>
<u>InstrumentScopeSecurityExchange(1616)</u>	<u>Empty</u>

8.3.2. BREACHING OF RISK LIMIT

This example shows the breaching action when an order exceeds the Maximum Buy Order Volume Limit. An ExecutionReport (8) follows a NewOrderSingle (D) in breach of a pre-trade check.

Example details:

- a) The Maximum Buy Order Volume Limit is assumed 100,000
- b) The LotSize is assumed to be 1
- c) An order is entered to buy 150,000 units



8.3.2.1. RM rejection codes which can appear in OrdRejReason (FIXTag: 103)

	Value
OrdRejReason (FIXTag: 103)	RiskLimitNotDefined (7000). This error appears in the case if no Pre-Trade limits have been defined.
	RiskMaximumOrderVolumeExceeded (7001)
	RiskMaximumOrderValueExceeded (7002)
	RiskOrderPriceCollarExceeded (7004)

8.3.3. RISK LIMIT INTERROGATION REQUEST 1A

This example shows the kinematics of requesting the value of a Maximum Buy Order Volume limit. The status and values of the risk limit are requested through a PartyRiskLimitsRequest (CL) and details are returned by a PartyRiskLimitsReport (CM).

The details of the PartyRiskLimitsDefinitionRequest (CS) limit definition used in this example are as follows:

1. The limit is defined by a clearing firm | LEIcode (FIXTag: 1658) = 11112222ClearingFirm
2. The limit applies to:
 - a) orders submitted by Exchange Member | LEI (FIXTag: 1691) = 88899999EXECUTINGFIRM
 - b) which concern clients | OrderCapacity (FIXTag: 528) = AGENCY (A)
 - c) who do not act as market makers | OrderAttribute (FIXTag: 2594) = Liquidity Provision Activity Order (2) and OrderAttributeValue (FIXTag: 2595) = No (N)
 - d) apply to shares listed on the Regulated Market | MICCode (FIXTag: 1616) = XWAR
3. the limit is for buy order volume | RiskLimitType (FIXTag: 1530) PER BUY ORDER VOLUME (301)
4. The value of the limit is set to 200,000 currency units | RiskLimitAmount (FIXTag: 1531) = 200000

FIX Trading Client
Risk Management
Application

WATS FIX
Trading Port

PartyRiskLimitsRequest(CL)

RiskLimitRequestID (1666) = 200

RiskLimitRequestType (1760) = Definitions (1)

RequestingPartyGrp

NoRequestingPartyIDs (1657) = 1

RequestingPartyID (1658) = 11112222CLEARINGFIRM

RequestingPartyIDSource (1659) = LEI (N)

RequestingPartyRole (1660) = Clearing Firm (4)

Parties

NoPartyID (453) = 1

PartyID (448) = 8889999EXECUTINGFIRM

PartyIDSource (447) = LEI (N)

PartyRole (452) = Executing Firm (1)

PartyRoleQualifier (2376) = Empty

RequestedRiskLimitTypesGrp

NoRequestedRiskLimitType (1668) = 1

OrderCapacity (528) = Agency (A)

OrderAttributeGrp

NoOrderAttributes (2593) = 1

OrderAttributeType (2594) = Liquidity Provision Activity Order (2)

OrderAttributeValue (2595) = No (N)

RiskLimitType (1530) = PER BUY ORDER VOLUME (301)

RiskInstrumentScopeGrp

NoRiskInstrumentScopes (1534) = 1

InstrumentScopeGrp

InstrumentScopeSecurityID (1538) = Empty

InstrumentScopeSecurityIDSource (1539) = Empty

MarketSegmentID (1300) = Empty

InstrumentScopeSecurityExchange (1616) = XWAR

PartyRiskLimitsReport(CM)

RiskLimitReportID (1667) = 300

RiskLimitRequestID (1666) = 200

RiskLimitRequestType (1760) = Definitions (1)

RequestResult (1511) = Valid request (0)

UnsolicitedIndicator (325) = Message is being sent as a result of a prior request (N)

[...]

RiskLimitTypesGrp

NoRiskLimitTypes (1592) = 1

RiskLimitType (1530) = PER BUY ORDER VOLUME (301)

RiskLimitAmount (1531) = 200000

RiskLimitAction (1767) = Empty

RiskLimitUtilizationAmount (1766) = Empty

RiskLimitUtilizationPercent (1765) = Empty

RiskWarningLevelGrp

NoRiskWarningLevels (1559) = Empty

RiskWarningLevelAction (1769) = Empty

RiskWarningLevelPercent (1560) = Empty

RiskWarningLevelName (1561) = Empty

RiskInstrumentScopeGrp

NoRiskInstrumentScopes (1534) = 1

InstrumentScopeGrp

InstrumentScopeSecurityID (1538) = Empty

InstrumentScopeSecurityIDSource (1539) = Empty

MarketSegmentID (1300) = Empty

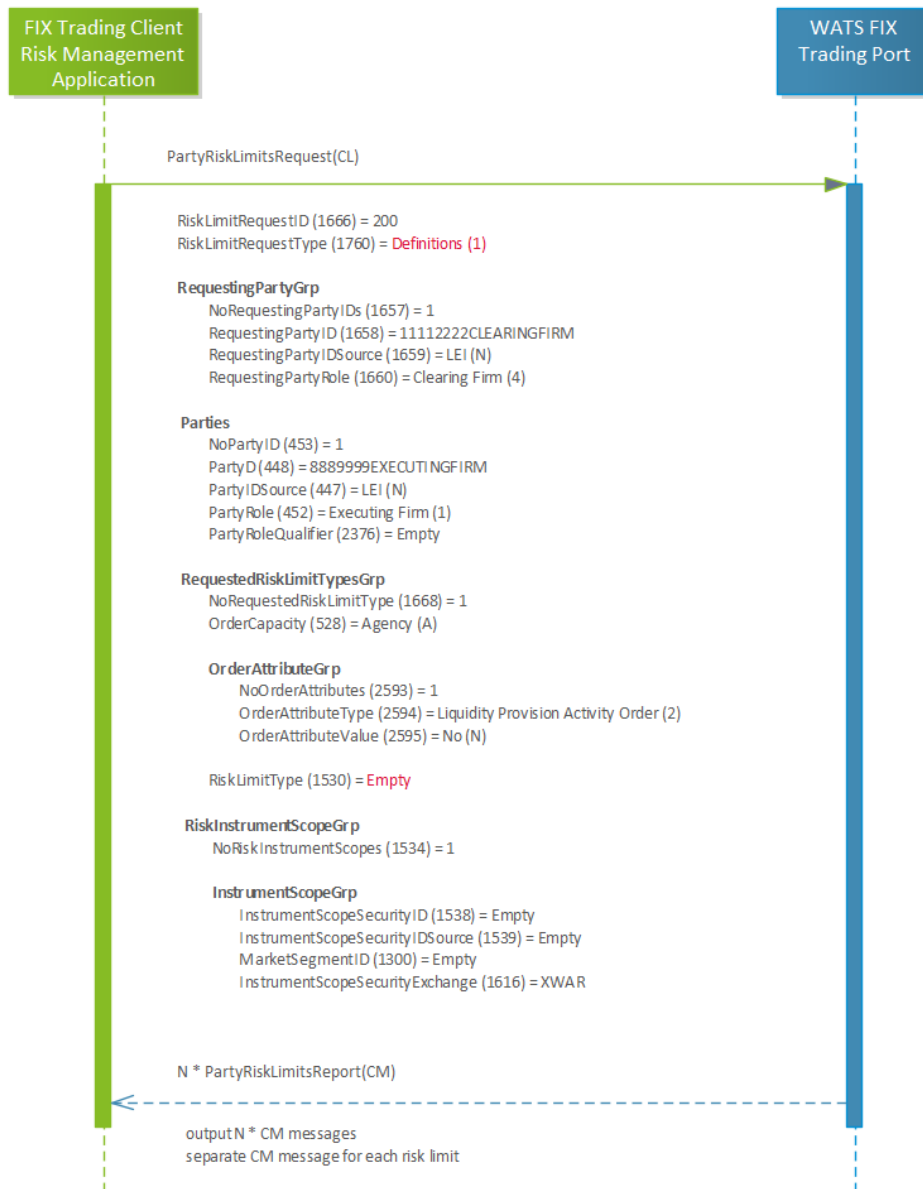
InstrumentScopeSecurityExchange (1616) = XWAR

8.3.4. RISK LIMIT INTERROGATION REQUEST 1B

This example shows the kinematics of requesting the value of any and all limits defined for Exchange Member with LEI 8889999EXECUTINGFIRM. The status and values of the risk limits are requested through a PartyRiskLimitsRequest (CL) and details are returned in a series of PartyRiskLimitsReport (CM) messages each containing details of a single limit.

The PartyRiskLimitsRequest (CL) request concerns the limits:

1. defined by clearing firm | LEIcode (FIXTag: 1658) = 11112222ClearingFirm
2. and which apply to:
 - a) orders submitted by Exchange Member | LEI (FIXTag: 1691) = 8889999EXECUTINGFIRM
 - b) which concern clients | OrderCapacity (FIXTag: 528) = AGENCY (A)
 - c) who do not act as market makers | OrderAttribute (FIXTag: 2594) = Liquidity Provision Activity Order (2) and OrderAttributeValue (FIXTag: 2595) = No (N)
 - d) apply to shares listed on the Regulated Market | MICCode (FIXTag: 1616) = XWAR



8.3.4.1. Examples of requests which can be constructed using PartyRiskLimitsRequest (CL)

	Type of Risk Limit	Type of Risk Limit And Market Structure	Type of Risk Limit And Market Structure and Exchange Member's type of activity
RequestedRiskLimitTypesGrp			
NoRequestedRiskLimitType (1668)	1	1	1
OrderCapacity (528)	A	A	Empty
OrderAttributeGrp			
NoOrderAttributes (2593)	1	1	Empty
OrderAttributeType (2594)	2	2	Empty
OrderAttributeValue (2595)	N	N	Empty
RiskLimitType (1530)	Empty	Empty	Empty
RiskInstrumentScopeGrp			
NoRiskInstrumentScopes (1534)	1	1	1
InstrumentScopeGrp			
InstrumentScopeSecurityID (1538)	Empty	Empty	Empty
InstrumentScopeSecurityIDSource (1539)	Empty	Empty	Empty
MarketSegmentID (1300)	Empty	Empty	Empty
InstrumentScopeSecurityexchange (1616)	XWAR	Empty	Empty

8.3.5. RISK LIMIT INTERROGATION REQUEST 2

This example shows the kinematics of requesting the value of a Total Traded Value limit and its utilization. The status and values of the risk limit are requested through a PartyRiskLimitsRequest (CL) and details are returned by a PartyRiskLimitsReport (CM). The report message shows a utilization of 5,000,000 (FIXTag: 1766) and the utilization percentage of 50% (FIXTag: 1766).

The details of the PartyRiskLimitsDefinitionRequest (CS) limit definition used to define this context are:

1. The limit is defined by a clearing firm | LEIcode (FIXTag: 1658) = 11112222ClearingFirm
2. The limit applies to:
 - a) orders submitted by Exchange Member | LEI (FIXTag: 1691) = 8889999EXECUTINGFIRM
 - b) which concern the institutional client with Short Code 100 | PartyDetailID (1691) = 100 and PartyDetailIDSource (FIXTag: 1692) = ShortCode (P)
 - c) who does not act as market maker | OrderAttribute (FIXTag: 2594) = Liquidity Provision Activity Order (2) and OrderAttributeValue (FIXTag: 2595) = No (N)
 - d) apply to shares listed on the Regulated Market | MICCode (FIXTag: 1616) = XWAR
3. the limit is for buy order volume | RiskLimitType (FIXTag: 1530) PER TOTAL TRADED VALUE (317)
4. The value of the limit is set to 10,000,000 | RiskLimitAmount (FIXTag: 1531) = 10,000,000

**FIX Trading Client
Risk Management
Application**
**WATS FIX
Trading Port**
PartyRiskLimitsRequest(CL)

RiskLimitRequestID (1666) = 400
 RiskLimitRequestType (1760) = **Definitions and utilisation (3)**

RequestingPartyGrp

NoRequestingPartyIDs (1657) = 1
 RequestingPartyID (1658) = 11112222CLEARINGFIRM
 RequestingPartyIDSource (1659) = LEI (N)
 RequestingPartyRole (1660) = Clearing Firm (4)

Parties

NoPartyID (453) = 2

---first repeating group--- exchange member

PartyD (448) = 8889999EXECUTINGFIRM
 PartyIDSource (447) = LEI (N)
 PartyRole (452) = Executing Firm (1)
 PartyRoleQualifier (2376) = Empty

---second repeating group--- institutional client

PartyD (448) = 100
 PartyIDSource (447) = ShortCode (P)
 PartyRole (452) = Client ID (3)
 PartyRoleQualifier (2376) = Firm or Legal Entity (23)

RequestedRiskLimitTypesGrp

NoRequestedRiskLimitType (1668) = 1
 OrderCapacity (528) = Agency (A)

OrderAttributeGrp

NoOrderAttributes (2593) = 1
 OrderAttributeType (2594) = Liquidity Provision Activity Order (2)
 OrderAttributeValue (2595) = No (N)

RiskLimitType (1530) = PER TOTAL TRADED VALUE (317)

RiskInstrumentScopeGrp

NoRiskInstrumentScopes (1534) = 1

InstrumentScopeGrp

InstrumentScopeSecurityID (1538) = Empty
 InstrumentScopeSecurityIDSource (1539) = Empty
 MarketSegmentID (1300) = Empty
 InstrumentScopeSecurityExchange (1616) = XWAR

PartyRiskLimitsReport(CM)

RiskLimitReportID (1667) = 500
 RiskLimitRequestID (1666) = 400
 RiskLimitRequestType (1760) = Definitions and utilisation (3)
 RequestResult (1511) = Valid request (0)
 UnsolicitedIndicator (325) = Message is being sent as a result of a prior request (N)

[...]

RiskLimitTypesGrp

NoRiskLimitTypes (1592) = 1
 RiskLimitType (1530) = PER TOTAL TRADED VALUE (317)
 RiskLimitAmount (1531) = 10000000
 RiskLimitAction (1767) = Empty
 RiskLimitUtilizationAmount (1766) = 5000000
 RiskLimitUtilizationPercent (1765) = 50

RiskWarningLevelGrp

NoRiskWarningLevels (1559) = Empty
 RiskWarningLevelAction (1769) = Empty
 RiskWarningLevelPercent (1560) = Empty
 RiskWarningLevelName (1561) = Empty

RiskInstrumentScopeGrp

NoRiskInstrumentScopes (1534) = 1

InstrumentScopeGrp

InstrumentScopeSecurityID (1538) = Empty
 InstrumentScopeSecurityIDSource (1539) = Empty
 MarketSegmentID (1300) = Empty
 InstrumentScopeSecurityExchange (1616) = XWAR

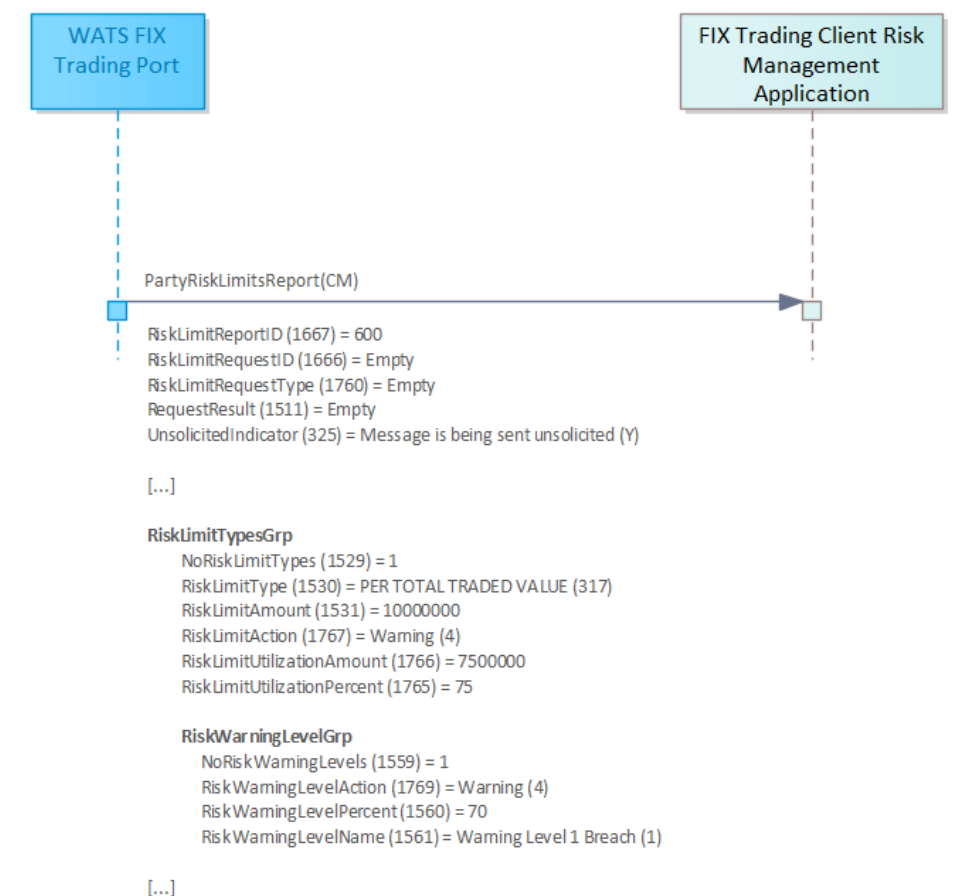
8.3.6. RISK LIMIT INTERROGATION REQUEST 3

This example shows the kinematics of WATS sending an unsolicited `PartyRiskLimitsReport (CM)` message following the breach of a warning level by a `NewOrderSingle (D)` order message. In this example, the warning level was set to 70%.

Step1: a `NewOrderSingle (D)` order message results in the breach of a warning level.

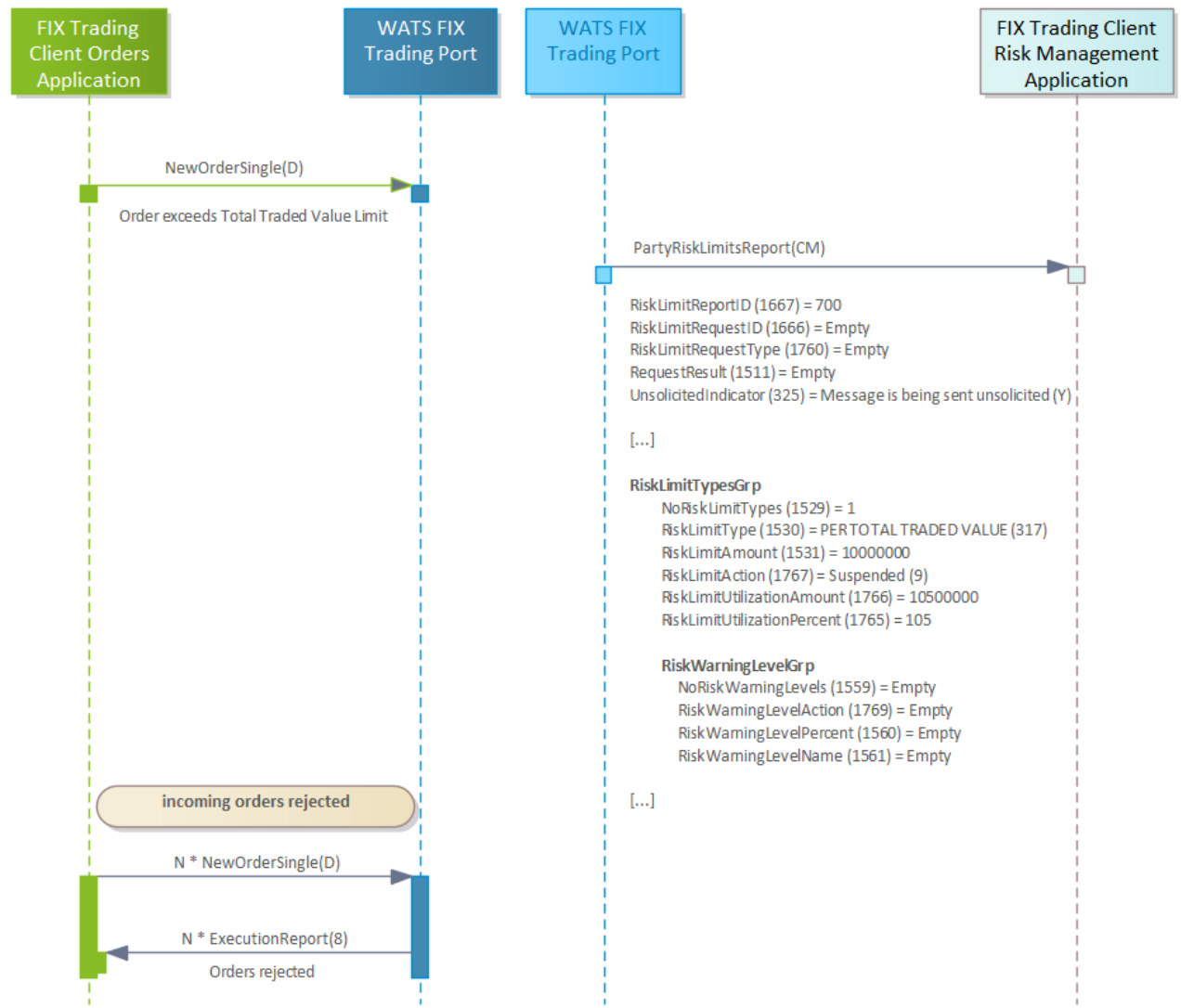


Step2: an unsolicited `PartyRiskLimitsReport (CM)` warns the user that a warning level was breached.



8.3.7. RISK LIMIT INTERROGATION REQUEST 4

This example shows the kinematics of WATS sending an unsolicited `PartyRiskLimitsReport (CM)` message in response to the breach of a post-trade risk limit by a `NewOrderSingle (D)` order message. In this example, the Total Traded Value limit was set to 10,000,000. The order is accepted and is followed by a `PartyRiskLimitsReport (CM)` message to the user. Subsequent orders are rejected by the RM subsystem.



8.4. ORDER IDENTIFICATION

GPW WATS supports various order types. The table below illustrates how these orders are distinguished are identified by the Risk Management subsystem.

Order Types	Native Trading Port	FIX Trading Port
	OrderAdd OrderModify	NewOrderSingle (D) OrderCancelReplace (G)
Limit Order	orderType=1	OrdType (40)=2 (Limit) Price (44)
Market Order (MO)	orderType=2	OrdType (40)=1 (Market)
Market to Limit (MTL)	orderType=3	OrdType (40)=K (Market with Left Over as Limit)
Stop Limit	orderType=5	OrdType (40)=4 (Stop Limit) Price (44) TriggerPrice (1102)
Stop Loss	orderType=6	OrdType (40)=3 (Stop Loss) TriggerPrice (1102)
Iceberg (Hidden) Order	orderType=4	OrdType (40)=2 (Limit) Price (44) DisplayQty(1138)

8.5. PRE-TRADE CHECKS AND ORDER TYPES

RM applies pre-trade controls to the following order types submitted to the system. Unpriced orders do not undergo pre-trade order price controls because they follow the market price.

Order Types	Order Price Control	Order Volume Control	Order Value Control
Limit Order	yes	yes	yes
Market Order (MO)	no	yes	yes
Market to Limit (MTL)	no	yes	yes
Stop Limit	yes	yes	yes
Stop Loss	no	yes	yes
Iceberg (Hidden) Order	yes	yes	yes